

Materials Engineer Test Reviewer

1. Scales for weighing aggregates and cement shall be accurate within ____% throughout the range of use. Scales shall be inspected and sealed as often as the Engineer may deem necessary to assure their continued accuracy.
Ans. 0.5 %
2. When cement is placed in contact with the aggregate, batches may be rejected unless mixed within ____ hours of such contact.
Ans. 1.5 hours
3. A sample from a bituminous mix weighs 1000 grams. At extraction, it was found out that the dry aggregate weighed 950 grams. What is the % of asphalt by weight of mix?
Ans. 5.0 %
4. The minimum penetration of the preservative with the surface of the timber piles shall be ____.
Ans. 200 mm
5. The contractor shall submit in writing a job-mix formula for the mixture supported by laboratory test data along with samples of resources of components and viscosity-temperature relationships to the Engineer for testing and approval at least ____.
Ans. three weeks
6. The concrete samples should be stored in a moist conditioned temperature within the range of ____ until the time of test.
Ans. 18°C to 24°C
7. This is used in soil stabilized road surface due to its high ____ property, that is, it absorbs and retains water.
Ans. Magnesium Chloride
8. The mass % wear of soil aggregate used in cement/lime proportions should not be more than ____.
Ans. 45 %
9. The uniform rate at which the movable head of the CBR machine travel is ____.
Ans. 0.10 in/min
10. The penetration test is an indication of the ____ of asphalt.
Ans. Consistency
11. Specification requires one (1) Quality test for every ____ of asphalt
 - a) 20 tons
 - b) 50 tons
 - c) 200 tons
 - d) 40 tons
12. Vibration shall operate a frequency of 8,300 to 9,600 impulses per minute under load at a maximum spacing of ____.
Ans. 60 cm
13. The specific gravity of Sodium Sulfate used in soundness test for aggregate should be within the range of ____.

Ans. 1.151 to 1.174

14. The specific gravity magnesium sulfate solution used in soundness test for aggregate should be within the range of ____.

Ans. 1.295 to 1.308

15. In checking the temperature of the mix, the thermometer should be inserted into the mix at least ____.

- | | |
|-------------|------------------|
| a) 2 inches | b) 4 inches |
| c) 6 inches | d) 6 millimeters |

16. ____ is applied on existing bit, or cement concrete surface with bituminous materials.

Ans. Bituminous Tack Coat

17. The solution used in the soundness test of aggregate shall be cooled to a temperature of $21 \pm 1^\circ\text{C}$ and maintain at that temperature for at least ____.

Ans. 18 hours

18. Proper compaction of the fill materials is very important, as it will ____.

Ans. All of the above

19. In which DPWH item is emulsified asphalt not suitable?

Ans. Item 301

20. Unconfined compressive strength test is usually performed on ____.

- | | |
|----------------|------------------|
| a) Sandy soils | b) Organic soils |
| c) Silty soils | d) Clayey soils |

21. Consolidation test is usually performed on ____ samples

- | | |
|----------------|-----------------|
| a) Disturbed | b) Consolidated |
| c) Undisturbed | d) both a & b |

22. A concrete core sample having a maximum height of less than ____ of its diameter before capping or height less than its diameter after capping shall not be tested.

- | | |
|---------|---------|
| a) 93 % | b) 95 % |
| c) 94 % | d) 96 % |

23. The cohesion and internal friction of soil is tested by;

Ans. All of the above

24. In concrete acceptance, core will be considered adequate if the average strength of the cores is equal to what?

Ans. 85% of specified strength & if no single core is less than 75% of specified strength

25. Before the cement is to be used, sampling must be done at least ____ before use.

Ans. one week

26. Asphalt cement shall not be used while it is foaming nor shall be heated at ____ at any time after delivery.

Ans. 159°

27. Moisture content of aggregate in bit mix shall not exceed ____ of the time it is introduced into the mixing unit.

Ans. 1 mass %
28. Batching tolerance for Sand and Aggregate is ____.

Ans. 3 %
29. As a Materials Engineer, what will you do with major honeycombed area that is considered as defective work?

Ans. Removed and replaced at the expense of the contractor
30. If the strength of control specimen does not meet the requirement and not feasible to obtain cores due to structural consideration, as a Materials Engineer what will be your decision?

Ans. Payment will be made at an adjusted price
31. After concrete has been placed, it is advisable to maintain fully wetted and in position for how many hours?

Ans. 72 hours
32. Rock shall be excavated to a depth ____ below sub grade within the limit of the roadbed

Ans. 150 mm
33. Minimum penetration of all piles shall be ____.

Ans. 3 meter
34. Piles shall not be driven within 7 meters of concrete less than ____ old.

Ans. 7 days
35. Paint contains ____ of pigment and ____ vehicle.

Ans. 21% & 79%
36. Minimum thickness of bedding materials beneath the pipe is ____.

Ans. 100 mm
37. Tolerance of gabion dimension is ____.

Ans. ± 3%
38. Minimum size of galvanized wire mesh is ____.

Ans. 2.2 mm
39. Minimum density of gabions is ____.

Ans. 1,400 kg/cu. m
40. Soundness loss of field gabions is ____.

Ans. 9% after 5 cycles
41. The heating temperature of reflectorized thermoplastic material during application is ____.

a) 130 ±7°C	b) 200 ±7°C
c) 150 ±7°C	d) 211 ±7°C
42. Submission of sample minimum quantity of material for asphalt cement is ____.

c) 115-150 psi/sec

d) 130-175 psi/sec

55. In concrete sampling paste shall be of normal consistency of ____ when the rod settles to a point below the original surface in 30 secs.

a) 10 ±1 mm

b) 15 ±1mm

c) 11 ±1 mm

d) 20 ±1 mm

56. What the curing temperature of a concrete sample?

Ans. 18-24°C

57. What is the wash loss of fine aggregates in the no. 200 sieve?

Ans. 3 %

58. Retarder can delay the setting time of concrete up to how many hours?

a) 1 hour

b) 4 hours

c) 2 hours

d) 8 hours

59. Super plasticizer can increase the strength of concrete up to how many percent?

a) 40-60 %

b) 50-70 %

c) 50-60 %

d) 60-70 %

60. ____ is the moisture in an air-dried soil

a) Hydrometer Moisture Content

b) Hygroscopic Moisture Content

c) Natural Moisture Content

d) Moisture Content

61. ____ is the tolerance of mix in bituminous.

a) 10°C

b) 15°C

c) 12°C

d) 18°C

62. ____ The number of measurement taken in a core sample for thickness determination of the finish asphalt pavement

a) 3

b) 5

c) 4

d) 8

63. In the absence of beam mold in site, cylinder mold is allowed by specification, what is the required strength?

a) 2,500 psi

b) 5,000 psi

c) 3,000 psi

d) 4,000 psi

64. Prime coat shall be left undisturbed for at least ____.

a) 6 hours

b) 24 hours

c) 12 hours

65. The first phase of rolling the spread bituminous mix is called breakdown rolling which is carried out by pneumatic roller, temperature requirement must be ____.

a) 200-225°F

b) 160-180°F

c) 190-199°F

d) 150-185°F

66. How many gradation ranges for hot plant mix bituminous pavement are there in the blue book?

a) 5

b) 7

c) 6

d) 8

67. What is the plasticity index of mineral filler?

Ans. not greater than 4

68. If no core had been taken at the end of each day of operation and the project has already been completed, a core shall be taken ____.

Ans. for every 75 l.m/lane

69. ____ is the slump of workable concrete if vibrated.

Ans. 10–40 mm

70. ____ Burnt product which is used by Gypsum and pulverized to form Portland Cement.

- | | | | |
|----|------------------|----|--------------|
| a) | Hydraulic cement | b) | Fly Ash |
| c) | Clinker | d) | Volcanic Ash |

71. When group index = 0, what is the grade of soil?

- | | | | |
|----|-----------|----|-----------|
| a) | weak soil | b) | good soil |
| c) | poor soil | | |

72. ____ is a process of breaking down of rocks into small pieces of various sizes that can range from large boulders to very small clay particles.

- | | | | |
|----|-----------------|----|------------|
| a) | Consolidation | b) | Weathering |
| c) | Unconsolidation | | |

73. ____ is the limit where further loss of water due to evaporation which will not result in reduction of volume.

- | | | | |
|----|---------------|----|-----------------|
| a) | Plastic limit | b) | Shrinkage limit |
| c) | Liquid limit | | |

74. ____ a classification of soil which 35% or less the particles pass through no. 200 sieve.

- | | | | |
|----|--------------------|----|----------------------|
| a) | Grain size soil | b) | Silty clay materials |
| c) | Granular materials | d) | Clayey materials |

75. ____ a classification of soil which 35% or more of the particles pass through no. 200 sieve

- | | | | |
|----|--------------------|----|----------------------|
| a) | Course Sand Soil | b) | Silty clay Materials |
| c) | Granular Materials | d) | Clayey Materials |

76. It evaluates the quality of soil as highway subgrade material.

- | | | | |
|----|------------------|----|--------------|
| a) | Atterberg Limit | b) | Group Index |
| c) | Plasticity Index | d) | Liquid Limit |

77. When soil has plasticity index value of 10, what is the criterion of the material?

- | | | | |
|----|--------------|----|--------|
| a) | Coarse grain | b) | Clayey |
| c) | Sandy | d) | Silty |

78. When plasticity index is equal

- | | | | |
|----|--------------|----|--------|
| a) | Coarse grain | b) | Clayey |
| c) | Sandy | d) | Silty |

79. ____ is an empirical number used in evaluation for G.I.

Ans. 0–20

80. It indicates good sub base materials when G.I. is equal to ____.

- | | | | |
|----|---|----|----|
| a) | 0 | b) | 10 |
|----|---|----|----|

c) 5

d) 20

81. It indicates very poor sub base materials when G.I. is equal to ____.

Ans. 20

82. ____ indicates the composition of soil.

Ans. Atterberg Limit

83. The higher the value of PI of the soil, the weaker will be the soil performance as a ____.

a) True
False

b) Depend
d) None of the above

c)

84. ____ the Plasticity Index of group of soil

Ans. 10

85. ____ is the minimum percent of A-6 group classification of soil.

Ans. 36 %

86. ____ is the maximum of G.I. of Silty Clay Materials

Ans. 20

87. When the value of G.I. is negative, what will be the soil performance?

a) Poor Subgrade
Very good subgrade

b) Good subgrade
d) None of the above

c)

88. A 25 grams fine grained soil can be dried in about ____.

a) 1 hour
c) 2 hours

b) 3 hours
d) 4 hours

89. A 25 grams sandy soil can be dried in about ____.

a) 1 hour
c) 2 hours

b) 3 hours
d) 4 hours

90. ____ is a type of soil in which most of the soil grains are of the same size.

a) Poorly graded
c) Well graded

b) Gap graded
d) None of the above

91. The soil grains are forced apart by the water added, thus the grain is reduced, what is the method used?

a) Lubrication
c) Wetting

b) Swelling
d) Saturation

92. Percent passing loose for coarse aggregates when tested for clay lump is ____.

a) 0.1 mass percent
c) 0.25 mass percent

b) 0.8 mass percent
d) 1.0 mass percent

93. The mass percent passing loose for coarse aggregates when tested in soft fragment is ____.

a) 0.10
c) 0.25

b) 3.5
d) 1.00

94. If slag is used, the density of coarse aggregate is ____.

Ans. 1,120 kg/cu. m

95. Mass percent loose for fine aggregates when tested in Clay lumps is ____.

a) 0.10	b) 0.80
c) 0.25	d) 1.0
96. ____ is a type of cement which shall be transferred to an elevated, air tight and weather proof bins storage.

a) Portland Cement	b) Bulk Cement
c) Pozzolan Cement	d) None of the above
97. The minimum thickness of conventional concrete as the surface course in monolithic pavement is ____.

a) 50 mm	b) 100 mm
c) 75 mm	d) 150 mm
98. ____ determines the resistance of the emulsion to break when mixed with mineral aggregate.

a) Viscosity Test	b) Spot test
c) Distillation Test	d) Cement Mixing
99. ____ is an instrument used for rapid in-site measurement of the structural properties of existing road pavement.

a) Pundit Ultra Sonic Concrete Tester	b) Dynamic Core Penetrometer
c) Falling Weight Deflectometer	d) Core Test
100. How many days do emulsified asphalt must be tested after delivery at temperature 40°F (4.5°C).

a) 10 days	b) 14 days
c) 12 days	d) 16 days
101. In bituminous mixture asphalt content is 5-8% and 92-95% aggregates, what will be the amount of additives?

a) not more than 2%	b) more than 2%
c) not more than 3%	d) more than 4%
102. ____ determines the effect of water on the cohesion of the mixture

a) Stability Test	b) Distillation
c) Immersion-Compression Test	d) Solubility
103. The distance of the sample by grooving before rotating the brass cup is ____.

a) $\frac{1}{4}$ "	b) $\frac{3}{4}$ "
c) $\frac{1}{2}$ "	d) 1"
104. The most common method of assessing strength and stiffness of both subgrade and pavement is ____.

a) FDT	b) SPT
c) CBR	d) None of the above
105. How many days do the specimen for CBR soaked?

a) 1 day	b) 3 days
c) 2 days	d) 4 days

106. When mixed at the site or in central mixing plant, the mixing time shall not be less than ____ unless mixer performance test prove adequate mixing of the concrete in a shorter time period.
- Ans. 50 seconds nor more than 90 seconds
107. The batches shall be so charged into the drum that a portion of the mixing water shall enter in advance of the cement and aggregate. The flow of water shall be uniform and all water should be in the drum by the end of the ____ of the mixing period
- Ans. 15 seconds
108. The time elapsed from the time water is added to the mix until the concrete is deposited at the site shall not exceed ____ minutes when concrete is hauled in the truck.
- Ans. 90 minutes
109. When concrete is delivered in truck mixers, additional water may be added to the batch and additional mixing performed to increase the slump to meet the specified requirement if permitted by the Engineer, provided all these operation are performed within ____ minutes after the initial mixing operation and the water and cement ratio is not exceeded
- Ans. 45 minutes
110. Concrete not placed within ____ minutes from the time the ingredients were charge into the mixing drum should not be use.
- Ans. 90 minutes
111. The completed pavement shall be accepted on a lot basis. A lot shall be considered as ____.
- Ans. 500 linear meter double lane
112. Each lot shall be divided into ____ equal segment and one core will be obtained from each segment in accordance with AASHTO T24.
- Ans. 5
113. Aggregate account for 92-95% of the weight of the bituminous mix while asphalts for 5-8% of the weight of the mix. The exact percentage to be used is determined by the ____.
- Ans. Trial Mix?
114. ____ is an indication that the asphalt is over heated
- Ans. Blue smoke
115. The penetrator test is an indication of the ____ of the asphalt
- Ans. Consistency
116. ____ is the most common used method in the design and evaluation of bituminous concrete mixes
- Ans. Marshall Stability Test
117. ____ determines the asphalt content in the bituminous mixes.
- Ans. Extraction Test
118. Cast in place concrete shall not be post-tensioned until at least ____ and until the compressive strength has reached the strength specified
- Ans. 10 days

119. ____ is the characteristic generally used for calculation of the volume occupied by the aggregate in various mixture containing aggregates including Portland Cement concrete, bituminous concrete, and other mixtures that are proportioned or analyzed on an absolute volume basis.
- Ans. Bulk Specific Gravity
120. ____ pertains to the relative density of the soil material making up the constituents particle not including the pore space within the particles that is accessible to water.
- Ans. Apparent Specific Gravity
121. Weight of sandy soil needed for Hydrometer test is ____.
- Ans. 100 grams
122. Weight of silty or clayey soil needed for Hydrometer test is ____.
- Ans. 50 grams
123. These are widely used to control the characteristics of soil which are to be incorporated in roadway.
- Ans. Liquid and Plastic Limits
124. ____ is defined as the highest moisture content of which the mass attains its minimum volume but continues to lose weight /mass.
- Ans. Shrinkage Limit
125. ____ is an apparatus used in determination of specific gravity of Hydraulic Cement.
- Ans. Le Chatterlier Flask
126. ____ is an equipment used in determination of soundness of Portland Cement
- Ans. Autoclave Machine
127. Asphalt shall be homogeneous, free from water, and shall not foam when heated to ____.
- Ans. 175°C (347°F)
128. ____ indicate the rate at which the asphalt emulsion will break when mix with aggregate .
- Ans. Cement Mixing Test of Asphalt
129. The lost in weight of rock used in gabions when tested to sodium sulfate, soundness loss shall not exceeds ____.
- Ans. 9%
130. Gabions dimensions are subject to a tolerance limit of ____.
- Ans. $\pm 3\%$
131. The pour point of concrete joint sealer shall be at least ____ lower than the safe heating temperature, which is the maximum temperature to which the material may be heated without exceeding the permitted flow.
- Ans. 11°C (20°F)
132. The penetration of concrete joint sealer, Hot Poured Elastic Type shall not exceed ____.
- Ans. 90
133. The CBR is generally selected at ____ penetration.

Ans. 2.54 mm (0.10 in)

134. If the ratio at 5.08 mm (0.21 in) penetration is greater, there should be a ____.

Ans. Rerun

135. If the check test gives a similar result, the ratio at 5.08 mm penetration shall be ____.

Ans. adopted or used

136. The penetration piston of CBR machine should have a uniform penetration at a rate of ____.

Ans. 1.30 mm/min

137. During soaking maintain the water level in the mold and the soaking tank is approximately ____ above the top of the specimen

Ans. 25.40 mm (1.0 in)

138. Soak the specimen for ____.

Ans. 96 hours (4days)

139. The diameter of penetration piston is ____.

Ans. 49.63 $\pm$ 0.23 mm (1935 cu m)

140. The temperature of the asphalt cement delivered to the mixer shall be as required to achieve a kinetic viscosity in the range of ____.

Ans. 150 to 300 cu m/s or 150 to 300 centistokes

141. ____ is the viscosity used to grade asphalt cement.

Ans. 149°F (60°C)

142. The coarse aggregate used in asphalt mix is ____.

Ans. Retained by 2.36mm (no 8 sieve)

143. The application temperature of emulsified asphalt range from ____.

Ans. 10-71°C

144. When the tar is used for seal coats, it shall be heated to free flowing but not to exceed ____.

Ans. 149°C?

145. ____ is used to determine the soil strength parameter.

Ans. Unconfined and Tri-Axial Compression Test

146. ____ is used to determine the settlement characteristic of soil.

Ans. Consolidation

147. The weight of disturbed samples from hard auger or split spoon shall be at least ____.

Ans. 1.0 kg

148. The undisturbed samples from Shelby Tube shall be at least ____.

Ans. 18" long

149. ____ determines the pile bearing capacity by loading one or more piles and measuring settlement under load.

Ans. Pile Load Test

150. _____ determines the in-site shear strength of soft to medium clay.

Ans. Vane Shear Test

151. _____ determines the soil bearing capacity by loading a steel plate usually 30 inches in diameter.

Ans. Plate Bearing Test

152. Laboratory which measures, examines, calibrate or otherwise determine the characteristic or performance materials or products.

Ans. Testing Laboratory

153. Standard technical procedure to determine one or more specified characteristic of material product is called _____.

Ans. Test Method

154. A document which prevents the test results and other information relevant to test is called _____.

Ans. Test Report

155. _____ is a document signed by the Director of BRS issued to Testing laboratory authorizing them to perform the required test in behalf of DPWH.

Ans. Certificate of Accreditation

156. This test has been developed to measure the loss of cohesion as a result of water action on the compacted bituminous mixture.

Ans. Film Stripping & Immersion-Compression Test

157. This mean porosity or perviousness of the mixture, a dense grade mix prevent water from entering the base through the surface.

Ans. Permeability

158. This is used in classifying liquid asphalt into grades. It is useful as a measure to control of consistency, particularly when run at the temperature of intended application

Ans. Viscosity Test

159. This test determines the total bitumen content of an asphaltic material, which is the actual cementing constituent of the asphalt.

Ans. Solubility Test

160. This is an accelerated test to show the loss of volatiles when the sample is heated at 163°C for five hours and the loss due to volatilization is determined.

Ans. Loss on Heating

161. This test classifies the different asphalt into types and the different types into grade.

Ans. Consistency Test

162. For liquid asphaltic materials, this test is run on the residue after distillation.

Ans. Quality Test

163. This is one of the most important engineering properties of soil. It is defined as its ability to resist sliding along internal surface within the mass.

Ans. Shearing Strength of Soil

164. This is an arbitrary measurement of consistency and is the basis of classification of all solid and semi-solid asphalt.

Ans. Penetration Test of Asphalt

165. This is a process to measure the consistency of liquid asphaltic materials to determine their fluidity or resistance to flow.

Ans. Viscosity Test

166. ____ was developed in order to differentiate bituminous materials with respect to hardening characteristic.

Ans. Thin Film Oven Test

167. Plant-mixed bituminous mixtures should be sampled in accordance with the Standard Method of Sampling Bituminous Mixture. A sample not less than ____ kg is taken and placed in sample bag that would insure against contamination. This is taken every 75 cu m or 130 tons of mix.

Ans. 10 kgs

168. The bearing blocks of the compression machine should be at least ____ greater than the diameter of the specimen.

Ans. 3.0 %

169. The bottom bearing block shall be at least ____ thick when new.

Ans. 1.0" (25.20 mm)

170. The bottom bearing block shall be at least ____ thick after any resurfacing operations.

Ans. 0.9" (22.50 mm)

171. The curing temperature of a specimen before testing is ____.

Ans. 18°C to 24°C

172. The apparatus used to determine the flash point of asphaltic material is called ____.

Ans. Cleveland Open Cup

173. The asphalt samples for ductility and penetration test should be subjected to water bath at a temperature of 25°C for ____.

Ans. 1 ½ hours

174. The concrete shall not be left exposed for more than ____ hour between stages of curing or during the curing period.

Ans. ½ hour

175. Another name for asphalt joint filler is ____.

- | | |
|-------------------|--------------------|
| a) asphalt cement | b) blown asphalt |
| c) rock asphalt | d) cutback asphalt |

176. In asphaltting work, there are ____ phase of rolling

- | | |
|------|------|
| a) 2 | b) 1 |
| c) 3 | d) 4 |

177. The ____ test determines the asphalt content in the bituminous mix.

- a) Grading
c) Bulk Specific Gravity
- b) Extraction
d) Absorption
178. In a bituminous mix, there are 3 kinds of aggregate. What kind of passes the no. 200 sieve?
- a) Coarse
c) Fine
- b) Mineral Filler
d) Admixture
179. Liquid asphalt is also called ____.
- a) Cutback asphalt
c) Emulsion
- b) Blown asphalt
d) Asphalt cement
180. The different test for soil classification are:
- a) compaction and FDT
c) Hydrometer test and moisture content
- b) Grading, LL and PL
181. The apparatus/equipment used in the liquid limit and plastic limit test is called ____.
- a) sieves and chisel
c) Atterberg Device and glass plate
- b) mortar and pestle
182. ____ is a method of laboratory sampling.
- a) Mixing
c) Using mechanical splitter/quartering
- b) Proportioning
183. In laboratory compaction test/moisture density relation test, what is the specification of rammer?
- a) 5.5 lb. (2.5 kg) with 12 in drop
b) 10 lb. (4.5 kg) in weight with 18 in drop
c) 8 lb. (3.6 kg) in weight with 16 in drop
184. What is the specification of mass present passing in sieve 0.075 mm (No. 200) grading "A" for aggregate base course?
- a) 20-14
c) 0-12
- b) 0-6
185. The number of specimen samples in testing concrete hollow block is ____.
- a) 4 pcs./20,000 units
b) 8 pcs./20,000 units
c) 6 pcs./20,000 units (3 for absorption test and 3 for strength test)
186. Is it really necessary to cure concrete specimen?
- a) Yes
c) None of the two
- b) No
187. The kind of test that determines the consistency of concrete is ____.
- a) Slump test
c) Specific Gravity
- b) Abrasion test
188. What is the dimension of concrete cylindrical sample?
- a) 6" dia. and 12" high
c) 4" dia and 8" high
- b) 8" dia. and 14 high
189. In the absence of concrete beam mold at the project site, are concrete cylindrical molds allowed by the specification to be used in sampler for item 311 - PCCP?
- a) Yes
b) Maybe

c) No

190. The minimum percent compaction for embankment is ____.

a) 92% b) 95%

c) 97%
191. The minimum percent compaction for base of the road is ____.

a) 100% b) 110%

c) 105%
192. The CBR value of aggregate base course is not less than ____.

a) 80% b) 90%

c) 85%
193. The CBR value of aggregate sub base course is not less than ____.

a) 25% b) 35%

c) 30%
194. The minimum compressive strength for class A concrete is ____.

a) 3,000 psi. (20.7 MN/sq. m.) b) 2,400 psi. (16.7 MN/sq. m.)

c) 5,000 psi. (37.7 MN/sq. m.)
195. What is the significance of determination of the tensile properties of reinforcing steel bars?

a) To determine the yield and tensile strength of the bar as well as its elongation, and is used to classify the bars into grade

b) To evaluate the ductile properties of the reinforcing bars

c) None of the above
196. A tie bar should be ____.

a) deformed bar b) round bar

c) flat bar
197. A dowel bar should be ____.

a) deformed bar b) round bar

c) flat bar
198. The mortar for masonry shall consist of ____.

a) one part Portland Cement and 2 parts of sand

b) one part Portland Cement and 3 parts of sand

c) one part Portland Cement and 4 parts of sand
199. The apparatus used to measure the thickness or height of compacted bituminous paving mixture.

a) Ruler b) Vernier Caliper

c) Steel Tape
200. It is used in calculating the unit weight of the mixture and the degree of compaction of the asphalt pavement.

a) Bulk Specific Gravity of compacted bituminous mixtures using saturated surface dry specimen.

b) Dry unit weight of compacted bituminous mixtures

c) Mass of specimen in water
201. The test which covers the quantitative determination of bitumen in hot mixed paving mixtures and pavement samples for specification acceptance, service evaluation, control and research.

202. The solvent used in extraction of bitumen is ____.

a) Diesel oil
b) Gasoline
c) Water

203. The percent of bitumen content is computed based on ____.

a) mass of raw sample
b) mass of dry aggregate
c) mass of asphalt

204. The method used to determine the grading of aggregate extracted from bituminous mixture is ____.

a) Stability Test
b) Viscosity
c) Mechanical analysis of extracted aggregate

205. The drying temperature of sample is ____.

a) $110 \pm 5^\circ\text{C}$
b) $100 \pm 5^\circ\text{C}$
c) 100°C

206. ____ is solid asphalt which is a basic constituent of all other asphalt called as hot asphalt or penetration grade asphalt.

a) Asphalt Cement
b) Emulsified Asphalt
c) Cutback Asphalt

207. ____ is a kind of liquid asphalt which is a mixture of asphalt cement and kerosene.

a) Medium Curing Cutback
b) Asphalt Cement
c) Rapid Curing Cutback

208. ____ is a kind of liquid asphalt which is a mixture of asphalt cement and oil.

a) Slow Curing Cutback
b) Emulsified Asphalt
c) Blown Asphalt

209. ____ is a kind of liquid asphalt which is a mixture of asphalt cement and gasoline

a) Rapid Curing Cutback
b) Blown Asphalt
c) Joint Filler

210. Asphalt is defined as ____.

a) A petroleum product
b) A dark brown to black cementitious materials whose consistency ranges from solid to semi-solid and whose main constituent is called bitumen which is either found in nature or as a residue in petroleum refining.
c) None of the above

211. It is a kind of liquid asphalt which is a mixture of asphalt cement and water with emulsified with pungent odor.

a) Emulsified Asphalt
b) Asphalt Cement
c) Rapid Curing Asphalt

212. Joint filler in concrete pavement is a ____.

a) Blown Asphalt
b) Hot asphalt
c) Liquid asphalt

213. A device used to measure the relative hardness of asphalt is ____.

- a) Penetrometer
c) Vicat Apparatus
- b) Spectrometer
214. A property of asphalt cement to stretch without breaking is called ____.
- a) Ductility
c) Tensile
- b) Durability
215. It is a point where the volatile components of asphalt will evaporate.
- a) Fire Point
c) Center Point
- b) Flash Point
216. It determines whether asphalt is over heated or not during the process of manufacture.
- a) Spot Test
c) Extraction Test
- b) Boil Test
217. An apparatus used to determine the specific gravity of asphalt is called ____.
- a) Metal Pycnometer
c) any bottle
- b) Glass Pycnometer
218. Consistency test for asphalt is called ____.
- a) Viscosity
c) Loss in ting
- b) Slump test
219. A process of volatilization test which measures the relative proportion of asphalt to oil.
- a) Distillation
c) Float Test
- b) Extraction
220. It is a period wherein the asphalt begins to melt.
- a) Softening Point
c) Fire Point
- b) Boiling Point
221. The sampling requirement of asphalt is ____.
- a) 1 sample for every 100 drums
b) 1 sample for every shipment
c) 1 sample for every 200 drums or 40 metric tons
222. The component of bituminous mixes which range from 92 to 96% is called ____.
- a) Bituminous Blended Aggregate
c) Sand
- b) Concrete Aggregate
223. ____ is a property of bituminous mixes in pavement to resist deformation or displacement due to impose load or repeated loading.
- a) Stability
c) Tensile
- b) Flexibility
224. ____ is a property of bituminous mixes in pavement to resist disintegration or deterioration due to the action of water, traffic and changing climate
- a) Durability
c) Ductility
- b) Flexibility
225. ____ is a property of bituminous mixes in pavement adjust itself to settlement
- a) Flexibility
c) Hardness
- b) Rigidity

226. These are the factor affecting the workability of bituminous mixes
- a) Gradation of aggregate, quality of asphalt, and temperature of mix
 - b) Surface texture and temperature mix
 - c) Quantity of aggregates and asphalt content
227. The dry molded asphalt sample in the immersion compression test is submerged in water for ____.
- a) 4 days
 - b) 4 hours
 - c) 2 days
228. The percent index retention is computed in the formula ____.
- a) $\text{Wet Stability/Dry Stability} \times 100$
 - b) $\text{Dry Stability/Wet Stability} \times 100$
 - c) $\text{Wet Stability} - \text{Dry Stability/Dry Stability} \times 100$
229. The sampling requirement of bituminous mixes is ____.
- a) 1 sample for every 75 cu. m.
 - b) 1 sample for every 150 cu. m.
 - c) 1 sample per delivery per day
230. The application of liquid asphaltic material on a prepared untreated base is ____.
- a) Prime Coating
 - b) Seal Coating
 - c) Tack Coating
231. The application of liquid asphaltic material to an existing concrete surface is ____.
- a) Tack Coating
 - b) Prime Coating
 - c) Seal Coating
232. The application of asphaltic material and aggregate to an existing surface for the purpose of sealing it against the infiltration of water or surface moisture is called ____.
- a) Seal Coating
 - b) Tack Coating
 - c) Prime Coating
233. The rate of application of prime coating using cutback asphalt is ____.
- a) 1-2 liters/sq. m
 - b) 0.5-1 liter/ sq. m
 - c) 2-3 liters/sq. m
234. The rate of application of tack coating using liquid or emulsified asphalt is ____.
- a) 0.2-0.7 li/sq.m
 - b) 0.1-0.5 li/sq. m
 - c) 2 li/sq. m.
235. The rate of application of seal coat using asphalt cement is ____.
- a) 0.9-1.8 li/sq. m
 - b) 10-11 li/sq. m
 - c) 8-9 li/sq. m
236. The rate of application of seal coat using cutback asphalt is ____.
- a) 1.5-3.0li/sq. m
 - b) 2.0-3.0 li/sq. m
 - c) 1-2 li/sq.m
237. The primary quality control mechanism for the production of asphalt tremies with high degree of uniformity that will satisfy job requirement is ____.
- a) Job mix formula
 - b) Quality control formula
 - c) Assurance
238. The objective of designing the asphalt mixture is determination of ____.

- a) Type and grade of asphalt, proportion of coarse, fine and mineral filler, if used and asphalt content.
 - b) To produce a desirable characteristic of pavement which are stable, durable, flexible and skid resistance.
 - c) All of the above
239. The percent asphalt content in hot laid bituminous mixture range from ____.
- a) 5-8%
 - b) 4-6%
 - c) 5-10%
240. The placing temperature for bituminous mixture measured in the truck prior to dumping is ____.
- a) at least 107°C
 - b) at least 120°C
 - c) at least 150C
241. The second phase of rolling the spread bituminous mixture is called breakdown rolling which is the one responsible to attain the maximum density. This rolling is carried out by the use of ____.
- a) pneumatic roller
 - b) steel roller
 - c) sheep foot roller
242. The significance of final rolling is to remove the roller marks to previous rolling in order to have a good finish pavement with a temperature of 160°F.
- a) True
 - b) False
 - c) Maybe
243. The proper way of rolling should begin at ____.
- a) the sides and proceeds longitudinally towards the roads center, each strip overlapping on half the roller width, gradually progressing the crown of the road
 - b) the center and proceeding longitudinally toward the side of the road
 - c) any part of the road provided each strip overlapping on half the roller width
244. What is the minimum degree of compaction of asphalt pavement?
- a) not lee than 97% of the density of the laboratory compacted specimen
 - b) not less than 100% of the density of the laboratory compacted specimen
 - c) not less than 85% of the density of the laboratory compacted specimen
245. What is the color of asphalt when it is overcook?
- a) yellowish brown
 - b) black
 - c) gray
246. ____ determines the thickness and density of compacted asphalt pavement.
- a) Core test
 - b) FDT
 - c) X-ray
247. The cored sample of asphalt pavement is measured at ____.
- a) approximately quarter points
 - b) three points
 - c) two points
248. Too much asphalt content in the bituminous mixes causes ____.
- a) bleeding
 - b) potholes
 - c) cracks
249. The wearing away of the pavement surface caused by the dislodging of aggregate particle is called ____.
- a) raveling
 - b) alligator crack

c) rutting

250. Interconnected cracks forming a series of small blocks resembling an alligator's skin or chicken wire is called ____.

a) alligator cracking
c) bleeding

b) reflection cracking

251. ____ is the type of cement to be used in the DPWH infrastructure project as indicated in the Blue Book.

a) Portland Cement Type I
c) Portland Cement Type II

b) Pozzolan Cement

252. If the fine aggregates are subjected to five cycles of sodium sulfate soundness test, the weighted loss shall not exceed ____.

a) 10 mass percent
c) 3 mass percent

b) 5 mass percent

253. The mass percent of wear of coarse aggregates for item 311 when tested by AASHTO 96 is not greater than ____.

a) 40%
c) 50%

b) 45%

254. The design of concrete mix specified by the blue book is based on ____.

a) Absolute Volume Method
c) American Method

b) Area Method

255. The minimum flexural strength requirement of beam sample when tested by third point loading method is ____.

Ans. 3.8 MPa

256. The minimum flexural strength requirement of beam sample when tested by midpoint method is ____.

Ans. 4.5 MPa

257. ____ are additives used in mixing concrete.

Ans. Admixtures

258. ____ determines the consistency of concrete.

Ans. Slump Test

259. Equipment used in the consolidation of fresh concrete.

Ans. Vibrator

260. The prescribed forms to be used in concrete paving are called ____.

Ans. Wooden

261. The initial setting time of Portland cement is not less than ____.

a) 30 min.
c) 90 min.

b) 60 min.

262. The final setting time of Portland cement Type I is ____.

a) 10 hrs.
c) 36 hrs.

b) 24 hrs.

263. The standard sand used in testing the mortar strength of Portland cement is called ____.

a) Ottawa Sand
c) River Sand

b) Beach Sand

264. The apparatus used in determining the fineness of Portland cement by air permeability is ____.

a) Blaine Air Permeability Apparatus
c) Gillmore

b) Sieve No. 200

265. A mixture of cement and water is called ____.

a) cement paste
c) concrete paste

b) mortar paste

266. The apparatus used to determine the initial and final setting of cement in the laboratory is ____.

a) Gillmore needles
c) Vicat apparatus

b) Bailey needle

267. The compressive strength of cement mortar sample in seven days is not less than ____.

a) 19.3 MN/sq. m
c) 24 MN/sq. m

b) 20 MN/sq. m

268. What is the appearance of the molded cement paste which fails to meet the autoclave expansion?

a) crumble
c) changes color

b) shiny

269. The dimension of beam sample for paving concrete is ____.

a) 6" x 6" x 21"
c) 4" x 4" x 218"

b) 6" x 6" x 24"

270. The critical number of days of curing concrete is ____.

a) first seven days
c) 28 days

b) first 10 days

271. The standard dimension of concrete cylindrical sample is ____.

a) 6" dia. and 12" tall
c) 6" dia. and 8" tall

b) 6" dia. and 6" tall

272. There are two layers in performing sampling of beam samples. How much number of blows is required per layer?

a) 63 blows
c) 50 blows

b) 75 blows

273. There are three layers in performing sampling of concrete cylinder. How many blows are required per layer?

a) 25 blows
c) 35 blows

b) 30 blows

274. The formula which determines the proportion of the mix which can meet the desired strength requirement of concrete is called ____.

a) design of concrete mix
c) Bernoulli's Formula

b) Boyle's Formula

275. The minimum testing requirement for every project is indicated in a duly approved ____.

- a) Quality Control Program
c) Work Schedule
- b) Program of Work
276. It is prescribed in each project based on estimated quantities and specifies the kind and number of test for each item of work.
- a) Minimum number of work
c) Work Schedule
- b) Minimum Testing Requirement
277. Embankment materials delivered at the job site was about 12,000 cu. m. The minimum test requirement calls for 1 - G, P, C for every 1,500 cu. m.
- a) 10 - G, P, C
c) 6 - G, P, C
- b) 8 - G, P, C
278. There is 1 - Q cement required for every ____.
- a) 4,000 bags or fraction thereof
c) 10,000 bags
- b) 2,000 bags or fraction thereof
279. There is 1 - QT RSB required for every ____.
- a) 20,000 kg or fraction thereof
c) 115,000 kg or fraction thereof
- b) 10,000 kg or fraction thereof
280. How many set of beam samples taken for every 75 cu. m. of concrete poured per day?
- a) one (1) set
c) three (3) sets
- b) two (2) sets
281. In a completed pavement, what is the minimum core samples required before payment is effected? (Assuming that the thickness is 23 cm.)
- a) Five holes per km per lane
c) P.E. Certificate
- b) One hole per 500m
282. In a Barangay road, what is the minimum requirement for thickness determination of its completed pavement?
- a) Five holes per km per lane
c) One hole
- b) P.E. Certificate
283. A quality test is one of the minimum requirements for item 200 - aggregate sub base course.
- a) True
c) optional
- b) False
284. There is no CBR test for item 300 - aggregate surface course.
- a) True
c) optional
- b) False
285. In item 311 and 405, grading is perform ____.
- a) for every 75 cu. m.
c) none
- b) for every 300 cu. m.
286. For pipe culverts and storm drains, the minimum test is 1 - Pipe for every 50 pipes. What is the alternative requirement?
- a) 1 - set consisting of 3 concrete cylinder samples for not more than 25 pipes cast in the field and 1 - inspection report for each size for not more than 25 pipes cast in the field.
b) inspection report is enough
c) Mill Certificate

287. The minimum test requirement for paint is ____.
- 1 - sample for every 100 cans or fraction thereof
 - 1 - sample for every 50 cans or fraction thereof
 - Mill Certificate
288. The minimum test requirement for water used in concrete for a questionable source is ____.
- P.E. Certificate
 - 1 - Quality Test
 - Inspection Report
289. Based on the maximum dry density obtained by laboratory test, the degree of compaction required for aggregate sub base should be at least ____.
- 95%
 - 100%
 - 98%
 - none of the above
290. Per our present Specification for Highways and Bridges including amendments, it is mandatory to open PCCP to traffic not later than ____.
- 7 days after pouring
 - 21 days after pouring
 - 14 days after pouring
 - 28 days after pouring
291. In casting reinforced concrete railings, the concrete used is ____.
- Class A1
 - Class C1
 - Class A2
 - any of the preceding classes
292. For efficient compaction, the embankment materials should at the time of rolling is ____.
- dry
 - at optimum moisture content
 - wet
 - at maximum moisture content
293. For a road base course the most preferable material is one with CBR of ____.
- 60
 - 50
 - 30
 - 80
294. Peat and muck are ____.
- fair subgrade spoil
 - unstable soil
 - stable soil
 - good subgrade soil
295. The slump test on concrete is a measure ____.
- strength
 - consistency
 - unit weight
 - durability
296. The fineness modulus is highest of ____.
- fine sand
 - very coarse sand
 - very fine sand
 - coarse sand
297. If concrete is continuously moist cured, it will gain strength beyond 180 days.
- True
 - it depends on curing temperature
 - False
298. The quality of paying concrete is usually measured by its ____.
- compressive
 - flexural
 - tensile
 - bond strength
299. Generally speaking, batching of concrete aggregates by volume is preferable than weight.

- c) they are the same

313. The standard compaction test on predominantly fine grained soils is performed on soil fraction passing sieve no. ____.

a) 200 b) 16
c) 40 d) 4

314. Which plasticity index would you prefer for the base course of PCCP?

a) less than 6 b) 10 to 12
c) 6 to 9 d) more than 12

315. A soil which is classified A-7 by the highway research board method will have a subgrade rating of ____.

a) excellent b) fair
c) good d) poor

316. The empirical number used in design is ____.

a) plasticity index b) group index
c) both of the preceding

317. Soil is fully saturated at optimum moisture content.

a) true b) false
c) it depends on void ratio

318. Water content of soil is generally based on ____.

a) total weight of soil b) weight of solids
c) total volume of soil d) unit weight test

319. Optimum water content of soil is determined by ____.

a) field density test b) compaction test
c) liquid limit test d) unit weight test

320. The particle size distribution of soil is determined by ____.

a) liquid limit granular b) sieve analysis test
c) plastic limit test d) any of the preceding tests

321. Which of the following are most suitable for construction of highway embankments?

a) granular b) plastic
c) fine-grained d) clayey

322. The particle size distribution of sandy gravel is determined by ____.

a) hydrometer test b) dilatancy test
c) sieve test d) any of the preceding tests

323. The best soil for a road subgrade would behave as a group index of ____.

a) less than 3 b) 11 to 15
c) 5 to 10 d) 16 to 20

324. The highest moisture content in the plastic state of fine grained soil is the ____.

a) liquid limit b) plastic limit

325. Which of the following is characterized as an unsuitable material?

a) soil with liquid limit exceeding 80

- b) soils with natural water content exceeding 100%
 - c) plasticity index exceeding 55
 - d) all of the above
326. Shrinkage factor test is ____.
- a) test to determine pay volume
 - b) test to determine strength characteristic
 - c) test to determine volume change
 - d) test to determine maximum density
327. Soil is fully saturated if the water content is ____.
- a) at optimum amount
 - b) more than 100%
 - c) 100%
 - d) none of the preceding
328. The density to which a soil can be compacted is related to ____.
- a) compactive effort
 - b) water content
 - c) both a and b
 - d) neither a or b
329. Specification require 1 quality test every ____ of asphalt mix.
- a) 40 ton
 - b) 50 tons
 - c) 130 tons
 - d) 150 tons
330. All asphalt are ____, they became harder as temperature decreases and softer as their temperature increases.
- a) thermoplastic
 - b) brittle
 - c) hydrophobic
 - d) none of the above
331. The viscosity of the asphalt during mixing and lay down is ____.
- a) viscosity at 135°C
 - b) viscosity at 60°C
 - c) viscosity at 125°C
 - d) viscosity at 163°C
332. The trial section for compaction for each type of fill materials and type of equipment to be used in the work should have an area of ____.
- a) 500 sq. m.
 - b) 600 sq. m.
 - c) 335 sq. m.
 - d) 400 sq. m.
333. Control slip shall have an area of approximately ____.
- a) 500 sq. m.
 - b) 600 sq. m.
 - c) 335 sq. m.
 - d) 400 sq. m.
334. Fine aggregate used in asphalt mix is the one ____.
- a) passing the no. 8 sieve
 - b) passing the no. 30 sieve
 - c) passing the no. 200 sieve
 - d) returned the no. 8 sieve
335. The corrugation produce by brooming is ____.
- a) 1.5 mm
 - b) 2.5 mm
 - c) 2.0 mm
 - d) 1.0 mm
336. The higher the ____, the lower the penetration.
- a) specific gravity
 - b) durability
 - c) viscosity
 - d) solubility
337. The required thickness of the thermoplastic paint in PCCP is ____.
- a) 3.2 to 4.8 mm
 - b) 3.3 to 3.8 mm

- c) 3.0 to 4.5 mm d) 3.2 to 4.0 mm

338. What percent of hydrated lime shall be added to the mixture during mixing operation of asphalt mix?

a) 0.5 to 1.0 mass % b) 0.75 to 1.2 mass %
c) 1.0 to 1.5 mass % d) 0.2 to 0.7 mass %

339. As Materials Engineer, what would you recommend in order to check the suitability and efficiency of the equipment to used prior to sub base construction for item 310, Bituminous Concrete Surface Course?

a) make a trial section b) make trial mix
c) establish control strips d) none of the above

340. Quality of factor – produced reinforced concrete pipes may be best established through what?

a) compressive strength test b) flexural strength test
c) core test d) crushing strength test

341. Result of Immersion–Compression test conducted on a bituminous mix sample are as follow:
Dry stability = 2,000 KPa, Wet stability = 1,000 KPa,
What is the resulting Index of Retained Strength?

a) 60% b) 50%
c) 55% d) 70%

342. In DPWH Contract Projects, Quality Control is the sole responsibility of ____.

a) DPWH project personnel b) Government's Materials Engineer
c) Contractor

343. As Materials Engineer, when is the right time to start sampling and testing of materials to be incorporated into the work?

a) before final payment b) before start of work items
c) before first partial payment

344. The Spot Test on asphalt is used to determine what?

a) cleanliness b) safe working temperature
c) overheating during manufacture d) viscosity of the mix

345. Under what climate conditions do we use higher penetration grade asphalt cement?

a) hot climate b) cold climate
c) rainy weather d) none of the above

346. What is the basis for rejection or acceptance of any construction materials?

a) Certificate of Billing b) Statement of work accomplished
c) Materials Test Report d) none of the above

347. Initial rolling of bituminous mixes shall be performed with a ____ roller.

a) vibrator b) sheep foot
c) pneumatic d) none of the above

348. In a bituminous mix sample, what is the % of asphalt by a) weight of aggregate and by b) weight of mix? Given the following data:

Weight of dry aggregates (after extraction)	=	950 grams
Weight of Mix	=	1010 grams

360. What control during the mixing and compaction of the bituminous pavement is of great significance in the resulting pavement?
- a) compaction
c) density
- b) grading of aggregates
d) temperature
361. In asphalt pavement construction, there are three phases of rolling; what kind of roller is required to be used for the final compaction and smoothing?
- a) pneumatic roller
c) tandem- smooth wheeled roller
- b) sheep foot roller
362. What is the maximum speed of the roller in compaction of asphalt pavement?
- a) 4 kph
c) 5 kph
- b) 8 kph
d) 6 kph
363. What shall be checked after final rolling?
- a) degree of compaction
c) quantity of aggregates
- b) optimum asphalt content
364. When is the completed asphalt pavement opened to traffic?
- a) when the asphalt has cooled to atmospheric temperature
b) when the asphalt has cooled to $\pm 10^{\circ}\text{C}$ of the spreading temperature
c) when asphalt is still hot
365. The apparatus used in determining the consistency of asphalt cement is ____.
- a) penetrometer
c) thermometer
- b) viscometer
366. How is the consistency of liquid asphalt determined?
- a) viscometer
c) penetrometer
- b) Tag open cup
367. The Optimum Asphalt Content obtained in Marshall Stability Test is the average asphalt content of the Maximum Stability, Maximum Bulk Specific Gravity, and ____.
- a) 4% Air Voids
c) 5% mineral content
- b) 2 mm flow
368. What test determines the safe working temperature of the asphalt?
- a) Ductility test
c) Stability Test
- b) Flash Point Test
369. ____ are usually tested for extraction.
- a) Asphalt Mixes
c) asphalt cement
- b) emulsions
370. Steel bars for the concrete reinforcement are tested for their ____ properties.
- a) tensile and bending
c) crushing
- b) compressive
371. What method is used in determining the strength of concrete pipes?
- a) third point loading method
c) three edge bearing method
- b) center point loading method
372. How many units shall be tested for strength, moisture and absorption in a lot of 120,000 CHB

Ans. 18 units

373. For 250 gallons of Reflectorized Traffic Paint, how many samples are required to be submitted for Quality Testing?

- a) 3 samples
c) 4 samples
- b) 2 samples

374. The bituminous mix should have a mass % of air voids within the range of ____.

- a) 2 to 3%
c) 3 to 5%
- b) 1 to 4%

375. Traffic paint shall be applied to the pavement at the rate of 0.33 li/sq. m. and shall dry sufficiently to be free it from cracking from ____.

Ans. 15 to 30 mins.

376. This is the only class of riprap that may consist of round natural stones.

Ans. Class A Riprap

377. The surface of riprap shall not vary from the theoretical surface by more than ____ mm at any point.

Ans. 100 mm.

378. Fine aggregates of masonry shall all passes the ____ mm sieve.

Ans. 2.36 mm (No. 8 sieve)

379. Mortar that is not used within _____ minutes after the water has been added shall be discarded.

Ans. 90 minutes

380. Maximum projection of rock faces beyond the pitch lines shall not be more than _____.

Ans. 50 mm.

381. In hot or dry weather, the masonry shall be satisfactory protected from the sun and shall be kept wet for a period at least _____ after completion.

Ans. Three (3) days

382. For pavement markings, the composition of white paint shall be ____ pigment and ____% of vehicle

Ans. 42-45 % pigment; 53-55% vehicle

383. The composition of yellow paint shall be _____ pigment and _____ vehicle.

Ans. 23% pigment and 77% vehicle

384. The ____ material shall be of such quality that it will not darken or become yellow when a thin section is exposed to sunlight.

Ans. Non-volatile material

385. ____ consist of beads of good quality, optically clear; lead free glass with not less than 90% reasonably spherical and free from flows.

Ans. Ballotini for reflective road paint

386. _____ is a reflector which shall be the short base type having a maximum base area of 180 mm x 140 mm.

Ans. Flash surface reflective stud

387. The pads shall be highly resilient and durable rubber reinforced with canvas and shall have a design life of at least ____.

Ans. Five (5) years

388. The reflectors shall have the minimum specific intensity value expressed as ____ of illumination.

Ans. Candle power per foot candle

389. The reflector shall support a vertical load of ____ when tested.

Ans. 10 KN

390. When raised profile type reflector is specified the approved ____ epoxy adhesive shall be used.

Ans. Two (2) parts

391. Expansion joint filler having a nominal thickness of less than ____ mm shall not be subject to a requirement for water absorption.

Ans. 9.5 mm

392. Expansion joint filler having a nominal thickness of less than ____ mm shall not be subject to a requirement for compaction.

Ans. 13 mm

393. Strip of the joint filler that do not conform to the specified dimension within the permissible variation of ____ mm thickness ____ in depth shall be reflected.

Ans. ± 1.6 to 0 thickness; ± 3 mm in depth

394. These are finely divided residue that result from the combustion of ground or powdered coal.

Ans. Fly Ash

395. A light spray application of asphalt to an existing pavement as a seal to inhibit raveling, or seal the surface or both.

Ans. Fog Seal

396. A light application of bituminous material for the express purpose of laying and bonding loose dust.

Ans. Dust Binder

397. The aggregate that is graded from the maximum size, down to and including filler, with the object of obtaining a bituminous mix with a controlled void content and high stability is called.

Ans. Dense-graded aggregate

398. The bituminous material used to fill and seal cracks in existing pavement is called ____.

Ans. Crack filler

399. The application of sprayed bituminous coatings not involving the use of aggregate is called ____.

Ans. Bituminous application

400. A mixture of bituminous material and mineral aggregate usually prepared in a conventional hot - mix plant or drum mixer at a temperature of not more than ____ and spread and compacted at the job site at a temperature above ____.

Ans. 260°F (127°C)/200°F (93°C)

401. ____ is an application of low viscosity bituminous material to an absorptive surface designed to penetrate, bond, and stabilize this existing surface and to promote adhesion between it and the new super-impose construction.
- Ans. Prime Coat
402. A uniform application of a mixture of emulsified asphalt, fine aggregate, mineral filler, and water to an existing pavement, single or multiple applications may be used.
- Ans. Slurry Seal
403. ____ is an application of bituminous material applied for an existing, relatively non-absorbent surface to provide a thorough bond between old and new surfacing.
- Ans. Tack Coat
404. ____ is a bituminous aggregate application to any type of road or pavement surface for the purpose of providing a wearing course, or surface seal or both.
- Ans. Treatment and Seals
405. ____ is a bituminous-sand application to an existing pavement to seal the surface and to function as a light - wearing course.
- Ans. Sand Seal
406. ____ is a spray application of bituminous material used to temporarily stabilize a recently seeded area. The bituminous material can be applied to the soil or to a straw or hay mulch as a tie down.
- Ans. Mulch treatment
407. ____ is a combination of bituminous material and aggregate that is physically mixed by mechanical means, spread on the job site and compacted.
- Ans. Bituminous Aggregate Mixtures
408. ____ is a combination of bituminous material and mineral aggregate applied at ambient temperature for use in patching holes, depression and distress area in existing pavement.
- Ans. Maintenance Mix
409. ____ is a pavement construction using essentially one-size aggregate which penetrated in place by an application of high viscosity bituminous material followed by an application of smaller size course aggregate and thoroughly rolled,
- Ans. Penetration Macadam
410. The glass beads used in traffic paint shall have a minimum of ____% true spheres.
- Ans. 70%
411. The glass beads shall have a minimum refractive index of ____.
- Ans. 1.50
412. The white thermoplastic material shall not exceed a yellowish index of ____.
- Ans. 0.12
413. The thermoplastic material shall have a maximum percent residue of ____ after heating at 8 ± 0.5 hours at $218 \pm 2^\circ\text{C}$
- Ans. 28%
414. The softening point of thermoplastic paint should be ____.

Ans. $102.5 \pm 9.5^{\circ}\text{C}$

415. What kind of water shall be used in laboratory test especially if it deals with chemical?

Ans. Distilled Water

416. Liquid limit brass cup with a sample is raised and allowed to drop sharply on the base through a height of what?

Ans. 10 mm.

417. How many rotations per second will the crank of a liquid limit device be rotated?

Ans. 2 rotations per seconds

418. What is the required size of calibrated sand used for FDT?

Ans. Any clean, dry, free- flowing, uncemented and passing # 10 and retained # 200 sieve

419. What is the degree of compaction of soil having a maximum dry density of 2,250 kg/cu. m. and wet density of soil from hole is equivalent to 2,335 kg/cu. m. and average moisture content of 8.85%?

Ans. $\% \text{ degree of compaction} = \frac{(\text{dry density}/\text{Max. dry density}) \times 100}{100} = \frac{[2,335/ (1+8.85/100)]}{2,250} \times 100 = \frac{(2,145.15/2,250) \times 100}{9.34\%}$

420. What is the standard diameter of an orifice in FDT sand cone?

Ans. $\frac{1}{2}$ " (12.7 mm)

421. Soil properties can be improved by ____.

- | | |
|------------------------|-----------------------------|
| a) Adequate compaction | b) Maximizing water content |
| c) Additional binder | d) Stabilization |

422. As construction progresses, you noticed that the materials delivered at the project site are different from the one tested and passed the requirement, as Materials Engineer what would you do?

- a) Authorize the use of the materials
- b) Conduct immediate retesting of the materials for verification purposes
- c) Reject materials
- d) Replace the materials

423. What is the test criterion for RCPC test with three (3) Edge Bearing Test?

- | | |
|-----------|-----------|
| a) 0.3 mm | b) 0.5 mm |
| c) 0.4 mm | d) 0.6 mm |

424. The Immersion - Compression test is prepared to determine ____.

- | | |
|---------------|----------------|
| a) ductility | b) stability |
| c) solubility | d) flexibility |

425. The use of sea water in reinforced concrete is not allowed due to ____.

- a) no effect
- b) cause of reduction in ultimate strength of concrete
- c) induce corrosion of R.S.B
- d) b and c

426. If the soil is properly compacted at MDD it will ____.
- | | | | |
|----|-----------------------------|----|----------------------------|
| a) | increase stability/strength | b) | minimize future settlement |
| c) | decrease permeability | d) | all of the above |
427. Asphalt construction, compaction must be completed before the mixture cools to ____.
- | | | | |
|----|--------------|----|--------------|
| a) | 165°F (60°C) | b) | 185°F (71°C) |
| b) | 175°F (65°C) | d) | 195°F (76°C) |
428. Which contractor is required by DPWH to own the minimum material testing equipment for pre-qualification purposes?
- | | | | |
|----|--------|----|------------------|
| a) | small | b) | large |
| c) | medium | d) | all of the above |
429. As percent asphalt increases in Marshall stability test, the % air voids ____.
- | | | | |
|----|-----------|----|------------------|
| a) | increases | b) | remains constant |
| c) | decreases | d) | a or c |
430. The aggregate component contributes to shear strength and stability of bituminous mixture. The desirable shape and surface texture of aggregate is ____.
- | | | | |
|----|----------------------------|----|---------------------------|
| a) | round and smooth surface | b) | round and rough texture |
| c) | angular and smooth surface | d) | angular and rough texture |
431. You can classify a soil sample in accordance to AASHTO through ____.
- | | | | |
|----|---------------------|----|-------------------|
| a) | Grain size analysis | b) | both a and c |
| c) | Atterberg Limit | d) | none of the above |
432. Who should take samples of materials and bring them to the laboratory for acceptance purposes?
- | | | | |
|----|---------------------------------|----|-------------------------------|
| a) | DPWH Materials Engineer | b) | Contractor's Project Engineer |
| c) | Contractor's Materials Engineer | d) | DPWH Project Engineer |
433. How do you determine the diameter of RSB?
- | | | | |
|----|--------------------|----|----------------|
| a) | by Vernier Caliper | b) | by unit weight |
|----|--------------------|----|----------------|
434. Cement which has been in storage for so long period that there is already doubt as to its quality should be ____.
- | | | | |
|----|-----------------------|----|-------------------|
| a) | retested prior to use | b) | authorized to use |
| c) | not to use | | |
435. In abrasion test, a sample of 5,000 grams was subjected to abrasion, a 2,000 grams sample was retained in no. 12 sieve after washing, what is the abrasion loss?
- | | | | |
|----|----|----|----|
| a) | 50 | b) | 60 |
| c) | 40 | d) | 30 |
436. Different brand of cement should not be allowed because ____.
- | | | | |
|----|-------------------------------|----|--------------------|
| a) | different in texture | b) | varies in strength |
| c) | different in color when dried | | |
437. An apparatus used for field density test is called ____.
- | | | | |
|----|----------------|----|-----------------------|
| a) | Sand cone | b) | Nuclear density meter |
| c) | Rubber balloon | d) | all of the above |
438. Thickness tolerance in the construction of sub- base course is ____.

- | | |
|----------------|----------------|
| a) ± 10 mm | b) ± 5 mm |
| c) ± 20 mm | d) ± 15 mm |

439. PH value of water used in concrete is ____.
- | | |
|-----------|------------|
| a) 4 to 6 | b) 6 to 10 |
| c) 6 to 8 | d) 3 to 5 |
440. In what item of work emulsified asphalt and cut back asphalt used?
- | | |
|---------------|---------------------------------------|
| a) Tack Coat | b) Seal Coat |
| c) Prime Coat | d) Bituminous Surface Course Hot-Laid |
441. What is the boundary between Coarse and Fine aggregates used in concreting?
- | | |
|--------------------------|------------------------|
| a) passing No. 3/8 sieve | b) passing No. 6 sieve |
| c) passing No. 4 sieve | d) passing No. 8 sieve |
442. The purpose of trial section is to check ____.
- | |
|--|
| a) suitability of materials |
| b) efficiency of the equipment |
| c) construction method proposed to be used |
| d) all of the above |
443. In FDT, the volume of hole is determined by what?
- | |
|---|
| a) Sand to fill hole over unit weight of dry sand |
| b) Total material taken from hole over volume of hole |
| c) Wet unit weight over $1 + MC/100$ |
| d) Dry unit weight over $MDD \times 100$ |
444. What is the average compressive strength of the load bearing concrete hollow blocks?
- | | |
|--------------|--------------|
| a) 1,000 psi | b) 600 psi |
| c) 800 psi | d) 1,200 psi |
445. The surface condition of tack coat before the next course should be applied is ____.
- | | |
|-----------|------------|
| a) wet | b) dry |
| c) sticky | d) plastic |
446. The contractor shall give the Engineer at least ____ days' notice prior to the beginning of work at the mill or shop where the metal structure has been manufactured or fabricated, so that the required inspection may be made.
- | | |
|-------|-------|
| a) 14 | b) 21 |
| c) 17 | d) 15 |
447. The request for recoring shall be made by the contractor within ____ weeks after the official result of the coring test has been released if the structure failed to meet strength requirement.
- | | |
|------------|--------------|
| a) one (1) | b) three (3) |
| b) two (2) | d) four (4) |
448. A revised design strength for concrete works was set forth from 14 days to 7 days for concrete pavement for national roads and ____ days for major interchanges and busy intersection thus minimizing occurrence of road congestion.
- | | |
|--------------|-------------|
| a) seven (7) | b) six (6) |
| b) three (3) | d) four (4) |
449. Compressive strength for mortar samples test at 3 days and 7 seven days can be considered for acceptance provided the strength of the samples at 7 days is not less than ____ MPa which is 85% of the compressive strength for 28 days

458. In PCCP coring, what is the specified thickness of an individual measurement that shall not be included in the average?

 - less than 20 mm
 - less than 25 mm
 - more than 20 mm
 - more than 25 mm

459. _____ is the wearing away of the pavement surface caused by the dislodging of aggregate particles.

 - Rutting
 - Shoving
 - Raveling
 - Corrugation

460. This is the stage in compaction in which all free air is expelled and replaced by water, if further pressure is applied to the soil mass, the tendency is to reduce its volume.

 - wetting stage
 - saturation stage
 - swelling stage
 - none of the above

461. In this stage, additional water replaces some of the entrapped air and the soil particles move closer together with a resulting increase in density.

 - wetting stage
 - lubricating stage
 - swelling stage
 - all of the above

462. All boring during soil exploration survey shall generally be carried to a depth of at least _____ meter below the proposed grade line.

 - one (1)
 - 0.30
 - two and half (2 ½)
 - 0.50

463. What is the factor that will influence the construction of stabilized road?

 - Depletion of the supply of economically available materials
 - High cost involved in the construction of the conventional types of road
 - If the road structure may be built by stages, and a properly designed road might function briefly as a wearing course
 - All of the above

464. What is the meaning of ASTM?

 - American Standard for Testing Materials
 - American Standard for Testing and Materials
 - American Society for Testing Materials
 - American Society for Testing and Materials

465. In sampling bituminous materials, the composite samples taken from a number of drums should have a size sufficient for testing and should not be less _____ as called for in the guidelines on the submission of samples.

 - 5 liters
 - 10 liters
 - 4 liters
 - 6 liters

466. For semi solid materials, samples should be taken at the center of the containers at least _____ mm. below the surface and 75 mm. from the sides with the aid of a clean hatcher or patty knife.

 - 100
 - 75
 - 85
 - 50

467. This is the most important volatilization test on liquid asphalt products, its gives information as to the actual amount of constituents remove at definite temperature.

 - Extraction test
 - Evaporation test
 - Distillation test
 - Softening test

468. This test indicates the purity of asphalt.

- | | |
|---------------------|---------------|
| a) Specific Gravity | b) Solubility |
| c) Viscosity | d) Ductility |

469. This is the most important factor which influences the stability of bituminous mixture.
- Gradation of mineral aggregate
 - Shape and surface texture of aggregate
 - Hardness of aggregate particles
 - All of the above
470. The principal factor that effects the flexibility of bituminous mixture is ____.
- the quality of bituminous binder
 - the quality of mineral filler
 - temperature susceptibility of the binder
 - all of the above
471. Concrete deposited in water shall be Class "A" concrete with a minimum cement factor of ____ per cubic meter of concrete.
- | | |
|-----------------------|-----------------------|
| a) 12 bags per cu. m. | b) 11 bags per cu. m. |
| c) 10 bags per cu. m. | d) 9 bags per cu. m. |
472. A tremie shall consist of a tube having a diameter of not less than ____ mm. constructed in sections having flanged couplings fitted with gasket with a hopper at the top.
- | | |
|--------|--------|
| a) 200 | b) 250 |
| c) 300 | d) 150 |
473. The fine aggregate used in stone masonry mortar shall all pass No. ____ sieve.
- | | |
|-------|--------|
| a) 40 | b) 30 |
| c) 8 | d) 3/8 |
474. The coarse aggregate used in Bituminous Concrete is ____.
- retained on a 2.36 mm. (No. 8 sieve)
 - retained on a 4.75 mm (No. 4 sieve)
 - retained on a 19.00 mm. (No. 3/4 sieve)
 - both a and b
475. As Materials Engineer, will you permit the use of salvage soil aggregate for lie stabilized road mix base course?
- Yes
 - It depends
 - No
 - Let the DPWH Project Engineer to decide
476. Is the accredited private testing laboratories owned by companies involved in the supply of construction materials for DPWH project be allowed to test their own products?
- | | |
|--------|------------------|
| a) Yes | b) It depends |
| c) No | d) Either a or b |
477. For small projects where the quantity of steel bars is less than 10,000 kgs. What may be required as a basis of acceptance?
- | | |
|---------------------|----------------------|
| a) P.E. Certificate | b) Inspection Report |
| c) Mill Certificate | d) All of the above |
478. Lumber is considered well-seasoned when its moisture content is between ____.
- | | |
|--------------|--------------|
| a) 10 to 16% | b) 10 to 14% |
| c) 14 to 18% | d) 15 to 20% |

- d) none of the above

489. When the embankment tends to soften in wet weather, and PCCP constructed of such material develop ruts under traffic and may shift and shove to develop a wash board surface, the plasticity index of such material is ____.

a) P.I. is too high
b) P.I. is too low
c) neither a & b
d) both a & b

490. What is the allowable smooth riding quality of PCCP?

a) 5.00 mm
b) 8.00 mm
c) 3.00 mm
d) 6.00 mm

491. What is the penetration grade of bitumen prime coat?

a) 60-70
b) 85-100
c) 120-150
d) none of the above

492. The spraying application temperature of all grades of asphalt cement should not be more than ____.

a) 211°C
b) 135°C
c) 159°C
d) 204°C

493. What is the rate of application of bituminous prime coat using cut back asphalt?

a) 0.5 to 0.7 liters/sq. m.
b) 1.0 to 1.1 liters/sq. m.
c) 1.0 to 2.0 liters/sq. m.
d) 1.10 to 2.0 liters/sq.m.

494. What is the weight boulders required for class A grouted riprap?

a) 10 to 25 kgs
b) 15 to 30 kgs.
c) 15 to 25 kgs.
d) 20 to 30 kgs.

495. Based on the maximum dry density obtained by laboratory test, the degree of compaction required for aggregate sub base should be at least ____.

a) 95%
b) 100%
c) 98%
d) 90%

496. Where to determine the amount of mixing water to use when compacting the soil in the field to attain maximum density.

a) Trial Section
b) Control Strip
c) Field Density Test
d) Laboratory Compaction Test

497. For soil classification purposes as in AASHTO M45, what is not required?

a) combined sieve and mechanical analysis
b) sieve analysis
c) the wash test
d) none of the above

498. In direct shear test, the soil is sheared to failure by moving one part of the soil sample relatively to the other part along a pre-determined plane. When several tests are made on a soil under different normal loads and the results are plotted in arithmetic scales, the resulting line is made in the form of ____.

a) straight line
b) parabola
c) hyperbola
d) semi-elliptical

499. In conducting soil exploration survey, what equipment you will use in obtaining undisturbed samples.

a) split spoon sampler
b) screw type soil auger sampler

- c) thin-walled sampler d) Post-hole auger sampler

500. Bituminous stabilization in the process of mixing intimately pulverized soil and bituminous material. What is the minimum bearing capacity (soaked CBR) of the mixture?

a) 80% b) 98%
c) 100% d) 95%

501. Portland cement stabilization road mix base course is an intimate mixture of pulverized soil, Portland cement and water which is thoroughly compacted while in a moist condition to form a structural road material when the cement hydrates and crystallizes. What is the mass percent wear of soil aggregate used in this stabilization?

a) 40% b) not more than 40%
c) 45% d) 50%

502. For a rich mixture of cement in concrete, what determines the flexural strength of the concrete?

a) Mortar strength b) both a & c
c) the strength of aggregate d) the water/cement ratio

503. This is a new method of soil stabilization being introduced to improve the properties of soil which is effective and at the same time cheaper.

a) Stabilization by the use of Geotextiles b) Cement stabilization
c) Bituminous stabilization d) Lime stabilization

504. As per guidelines on submission of samples, laboratory trial mixes should be made on aggregates which are proposed tube used for the first time, in this case what is the weight of fine aggregates to be submitted for testing?

a) 100 kgs. b) 50 kgs.
c) 70 kgs. d) 80 kgs.

505. The phosphorous content of reinforcing steel bars bigger than 9.0 mm shall not exceed ____.

a) 0.18% b) 0.06%
c) 0.08% d) 0.05%

506. The permissible variation of reinforcing steel bars bigger than 9.0 mm. shall not exceed ____.

a) 10% b) 5%
c) 8% d) 6%

507. A type of cement that sets and hardens by chemical interaction with water and is capable of doing so underwater.

a) Portland Cement b) Hydraulic Cement
c) Air-Retaining Portland Cement d) All of the above

508. The spacing center to center of circumferential reinforcement of all pipes shall not exceed its wall thickness and shall in no case exceed ____.

a) 150 mm b) 200 mm
c) 100 mm d) 250 mm

509. Oversanded and undersanded concrete mixtures require more water in order to be workable but will only result to ____.

a) good surface texture b) water tightens
c) weak strength d) all of the above

510. For every lean mixes or low cement content of the mixture, what determines the flexural strength of the concrete?

a) strength of aggregate c) mortar strength

b) water cement ratio

d) All of the above

511. This is a method of determining in-place density of soil that can be used satisfactorily in moist, cohesive fine grained material only. The sample is obtained by forcing a small thin walled cylinder into the material with a driving head and hammer or other suitable apparatus.

Ans. Drive Cylinder Method

512. This method of test is intended to determine the density of cohesive soil and stabilized soil by measuring the weight and volume of undisturbed samples.

Ans. Block, Chunk or Core Sampling

513. This method covers the determination of the density in-place of compacted or firmly bonded soil by using a flexible membrane and liquid that can be used to measure the volume of a test hole soil with an accuracy within 1.0%.

Ans. By Rubber Balloon Method

514. This method covers the determination of the total or wet density of soil and soil aggregate in-place by the attenuation of gamma rays, where the gamma detector, or both, remains at or near the surface.

Ans. Nuclear Method

515. What are the three major groups of soil?

Ans. 1) granular
2) fine grained
3) organic

516. What are the common laboratory tests required for soil?

Ans. a) sieve analysis
b) plastic and liquid limit test
c) compaction test (moisture-density relation)
d) CBR
e) Abrasion Test (Mass percent of Wear)
f) Unit weight
g) Moisture Content

517. What is the common field test?

Ans. To determine the degree of compaction of soil or it is a control test in embankment construction to ensure adequate compaction

518. What is moisture content?

Ans. It is the amount of water present in the soil

519. What is the different between natural moisture content with hygroscopic moisture content?

Ans. Natural moisture content is the moisture of undisturbed soil sample while hygroscopic moisture content is the moisture of air-dried soil sample.

520. What is the formula used for moisture content determination?

Ans. $\text{Moisture Content} = (\text{wt. of water} / \text{wt. of oven dried sample}) \times 100$

521. Weighing hot sample is not advisable as it affects the accuracy of result.

Ans. True

522. What are the methods use in reducing samples from the field to testing size?

Ans. 1) sample splitter

2) quartering

523. What is the required temperature for oven drying the soil sample?

Ans. $110^{\circ}\text{C} \pm 5^{\circ}\text{C}$

524. How can you determine if the soil sample is already oven dried?

Ans. dried if it reached its constant weight

525. How to prepare or to come up with air-dried soil sample?

Ans. air dries the sample under the heat of the sun

526. If impossible to air-dry the sample under the heat of the sun what alternative should be used?

Ans. oven dry the sample at 60°C

527. Sample for compaction test should pass to what sieve?

Ans. method A & B use sample passing sieve #4; method C & D use sample passing #3/4

528. What is the difference between T-99 and T-180 compaction test method?

Ans. T-99 uses 2.5kg.(5.5lbs.) rammer with a drop of 12" height, while T-180 uses 4.54kg.(10 lbs.) rammer with a drop of 18" height

529. What is the particle size analysis?

Ans. It is the determination of particle size distribution in soil by sieve, hydrometer, or combined analysis

530. What are the other terms or particles size analysis?

Ans. 1) sieve analysis
2) mechanical analysis
3) grading test/analysis

531. What is the liquid limit?

Ans. it is the lowest moisture content at which the soil will flow upon the application of a very small shearing force.

532. The liquid limit is expressed as the moisture content corresponding to what number of blows?

Ans. 25 blows

533. What tests are used in soil classification?

Ans. 1) grading or sieve analysis
2) plastic and liquid limit

534. What is plastic limit?

Ans. It is the minimum moisture content at which the soil can be readily molded without breaking or crumbling

535. What kind of water shall be used in laboratory test especially if it deals with chemicals?

Ans. Distilled water

536. Liquid limit brass cup with sample is raised allowed to drop sharply on the base through a height of what?

Ans. 10 mm.

537. How many rotations per second will the crank of a liquid limit device be rotated?
 Ans. 2 rotations per second
538. In doing a plastic limit test, to what diameter shall the soil thread began to break?
 Ans. 3.2 mm. (1/8")
539. What is the formula for plastic limit?
 Ans. $PL = (W_a - W_b)/W_b \times 100$
 Where: W_a = original wt of crumbled soil thread
 W_b = wt of oven dried crumbled soil
540. What is Plastic Index (PI)?
 Ans. It is the difference between the liquid limit and the plastic limit;
 $PI = LL - PL$
541. What is specific gravity?
 Ans. It is a gravimetric-volumetric relationship in soils (or defined as the ratio of the weight in air of an equal volume of water at a stated temperature)
542. What is the approximate area for density control strip?
 Ans. 335 sq. m.
543. What is the maximum depth of roadway embankment per horizontal layer in loose measurement?
 Ans. 200 mm (8")
544. What item of work is embankment?
 Ans. Item 104
545. What is the kind of material used for item 200?
 Ans. Aggregate sub base course
546. What is the required particle size for selected borrow for topping under item 104 (embankment)?
 Ans. All particle size will pass a sieve 75 mm. or 3" square opening and not more than 15% will pass 0.075 mm. (No. 20) AASHTO 11
547. P.I. requirement for item 104 (selected common borrow) is _____.
 Ans. 6 maximum
548. L.L. requirement for item 104 (selected common borrow) is _____.
 Ans. 30 maximum
549. What is the LL requirement for item 200 (aggregate sub base course) passing sieve # 40?
 Ans. 12 maximum
550. Coarse portion retained on sieve # 10 (2.00 mm.) of item 200 shall have a mass percent of wear by Los Angeles Abrasion test of _____.
 Ans. 50% maximum
551. The required CBR value for item 300?

Ans. 25% minimum (AASHTO T 180 method D)

552. What is the minimum percent field density or degree of compaction for item 104 (embankment)?

Ans. 95% minimum

553. What is the minimum degree of compaction for item 200 and 201?

Ans. 100% minimum

554. What is the highest point in moisture density curve of the moisture density relation test or compaction test of soil represent?

Ans. MDD (Maximum Dry Density) and OMC (Optimum Moisture Content)

555. What is the required size of calibrated sand used for FDT?

Ans. Any clean, dry, free-flowing uncemented sand passing # 10 and retained # 200 sieves

556. What is the degree of compaction of soil having a maximum dry density 2,250kg/cu. m. and wet density of soil from hole equivalent to 2,335kg/cu.m. and an average moisture content of 8.85%?

Ans. % Degree of Compaction = (Dry density/Max. dry density) x 100

Where ;
Dry density = Wet density/(1+MC/100)
Wet density = wt of soil filled in a hole/volume of hole
% degree of compaction = {(2,335)/[1+(8.85/100)] x 100}/2,250
= (2,145.15/2,250) x 100
= 95.34%

557. Is the soil for No.45 passed the item 104 requirement for % degree of compaction?

Ans. G or Grading = 2
P or Plasticity = 2
C or Compaction = 2
D or Density = three in-site density test/500 sq. m.

558. What are the test and number of test needed for 2,960 cu. m. Aggregate Base Course?

Ans. sand cone, jug (at least 4 liters), guide plate, moisture cans, sand, weighing scale, oven with temperature control, chisel or digging tools, plastic bags and labeling materials (tag number)

559. What is the standard diameter of an orifice in FDT sand cone?

Ans. ½ inch (12.7 mm.)

560. How many hours do a fine aggregates tested for specific gravity and absorption be soaked in water?

Ans. 15 to 19 hrs.

561. How will you determine if the fine aggregates reach the saturated surface dry condition?

Ans. It is determine by the use of cone test for surface moisture and if the molded shape of fine aggregate slump slightly it indicates that it has reached a surface-dry condition.

562. What are the tests required for concrete aggregates?

Ans. 1) Fine Agg: Grading, specific gravity, absorption, dry unit wt.
2) Coarse Agg: grading, specific gravity, absorption, dry unit wt, and abrasion

Note: soundness is also tested as per request

563. What is the maximum mass % of wear by Los Angeles Abrasion Test?
Ans. 40% maximum
564. What is the required speed of Los Angeles Abrasion Machine per minute?
Ans. 30 to 33 rpm
565. What is the size or diameter and weight of cast-iron spheres use in abrasion?
Ans. Approximately 46.8 mm dia. And weighing between 390-455 grams
566. What is the required total wt. of sample for abrasion test of coarse aggregate grading A with 12 number of sphere?
Ans. 5,000 grams $\pm$ 25 grams
567. What is the significance of abrasion test?
Ans. It evaluate the structural strength of coarse aggregates - gives an indication of quality as determined by resistance to impact and wear - it determines whether the aggregate will have degradation during traffic or rolling
568. Quality test for cement is represented by how many bags of cement?
Ans. 2,000 bags or fraction thereof
569. What is the sieve use in sieving materials from Abrasion Machine?
Ans. Sieve # 12 (1.70 mm.)
570. Testing machine should be calibrated once every six (6) months if possible but if it not possible what is the required time should the machine be calibrated?
Ans. Once a year
571. What is the percent tolerance of error in calibrating the machine?
Ans. $\pm$ 1%
572. What do you call a sample for concrete pavement?
Ans. Concrete beam sample
573. How many samples should be taken for every 75 cu. m. of fresh concrete?
Ans. 3 pcs. sample or 1 set sample
574. What do you call the sample for structural concrete?
Ans. Concrete cylinder sample
575. What is the test use to determine the consistency of concrete?
Ans. Slump test
576. How to compute for the slump of concrete?
Ans. Slump = Ht. of slump cone - Ht. of fresh concrete after subsidence
577. How to compute for the compressive strength of concrete cylinder sample?
Ans. $CS = P/A$

where: P = loads, lbs.
 A = area, sq. m.
 A = $[rD^2]/4$
 = 28.27 sq. m

578. At what age should a concrete beam sample be tested for flexural test?

Ans. 4 days

579. How to compute the flexural strength for third point loading test?

Ans. R = $(PL)/(bxdxd)$
 = $P(18'')/(6'' \times 6'' \times 6'')$
 = $P/12$

where: R = modulus of rupture
 P = load in lbs. or tons
 L = span length in inches
 b = base in inches
 d = depth in inches

580. What is the minimum flexural strength for third point loading test?

Ans. 3.8 MPa (550 psi)

581. What is the minimum flexural strength for center point loading test?

Ans. 4.5 MPa (650 psi)

582. How to compute for flexural strength at center point loading?

Ans. R = $(3PL)/(b)(d^2)$
 = $3P(18)/(2 \times 9 \times 6 \times 6)$

583. Which part of the batch for fresh concrete should the sample be taken from stationary mixers, revolving drum truck mixers or agitators?

Ans. two or more regular intervals during discharge of middle portion of the batch

584. What is the size of tamping rod use in sampling concrete?

Ans. 16 mm. (5/8 inches) diameter and 610 mm. (24 inches) long with the tamping end rounded to a hemispherical tip of the same dia. as the rod

585. Concrete Masonry (hollow blocks) if subject to test, how many sample is required?

Ans. 6 pcs./10,000 units 12 pcs. if more than 10,000 units

586. What is the strength requirement for load bearing concrete masonry units?

Ans. Individual = 5.5 MPa (800 psi) minimum
 Average (for 3 samples) = 4.14 MPa (600 psi) minimum

587. What is the significance of testing reinforcing steel bars?

Ans. to determine the yield and tensile strength of the bar as well as its elongation and is used to classify the bars into grade

588. What is the required length of RSB for testing?

Ans. 1 meter/10,000 kgs./size/shipment

589. What is the significance of bending test for RSB?

Ans. to evaluate the ductile properties of RSB

590. What is the required size of test specimen for G.I. sheets?
Ans. 60 mm. x 60 mm. (3pcs. from one sheet/100 sheets)
591. What is the required sample for paint?
Ans. 1 can (gal. or pail)/100 cans (gal. or pail)
592. What kind of paint had reflectance in the form of beads?
Ans. Reflectorized Traffic Paint
593. What is the common component of house paint?
Ans. pigment and vehicle
594. What is this construction material that is dark brown to black cementitious material in which the predominating constituent is bitumen?
Ans. Asphalt
595. What asphalt is commonly called hot asphalt?
Ans. Asphalt cement
596. What is liquid asphalt?
Ans. Cut-back asphalt and Emulsified Asphalt
597. What is the use of blown asphalt or hard asphalt?
Ans. Use as joint filler or water proofing
598. SS-1 or SS-1h is what kind of asphalt?
Ans. Slow Setting Emulsified Asphalt
599. What are the three (3) kinds of Emulsified Asphalt?
Ans. Rapid Setting, Medium Setting and Slow Setting
600. What asphalt contains water?
Ans. Emulsified Asphalt
601. What is the penetration grade for blown asphalt?
Ans. 0 to 30
602. What is the ideal penetration grade of asphalt use in Tropical Region?
Ans. 85 to 100 (but some uses 60 to 70)
603. Tack coat is used in an existing bituminous or concrete pavement.
Ans. True
604. What is the rate of application of bituminous tack coat?
Ans. 0.2 to 0.7 liter/ sq. m.
605. What is the rate of application of bituminous prime coat?
Ans. 1 to 2 liters/sq.m.
606. For how many hours or how long shall a bituminous prime coat left undisturbed?

Ans. 24 hrs.

607. What kind of asphalt is used as prime coat?

Ans. Cut-back asphalt (medium curing or rapid curing)

608. Bituminous prime coat is used to treat surface of base course.

Ans. True

609. No tack coat or prime coat shall be applied when the weather is foggy or rainy.

Ans. True

610. What kind of asphalt is used as tack coat?

Ans. cut-back asphalt (rapid curing) or emulsified asphalt

611. Where should the bituminous seal coat be used?

Ans. on an existing bituminous surface course with or without an application of aggregates

612. If concrete cylinder sample for RCP is not available for compression test one (1) whole piece RCP for every 50 pcs. RCP will be submitted for quality test.

Ans. True

613. One set (3 pcs) concrete cylinder sample shall be taken for every how many pieces of RCP?

Ans. 25 pcs.

614. In submitting samples for testing, what particular form is being accomplished?

Ans. sample card

615. As a Materials Engineer of the project, what is the report accomplished every week?

Ans. Quality Control Assurance Report

616. What do you call the report to be submitted at end of the month?

Ans. Monthly Materials Report

617. How to determine the proportion of cement, water, fine aggregates and coarse aggregates?

Ans. by means of Concrete Design Mix

618. How to determine the thickness of pavement?

Ans. by means of concrete core

619. How to determine the asphalt content of bituminous mix?

Ans. by means of extraction test

620. In an extraction test, aside from the asphalt what other materials being extracted?

Ans. Aggregates or composite aggregates (minerals of fine & coarse aggregates)

621. What is being determined in a core sample from asphalt pavement?

Ans. thickness of pavement and density of pavement

622. All records regarding quality control such as accomplishment, daily activities, weather etc. are recorded in a ____.

Ans. Materials Log Book

623. The standard penetration test (SPT) is an in-site test that measures what?

Ans. depth of soil layer

624. Which of the following bituminous material is best recommended for sealing weakened plane joints?

Ans. Joint Filler Asphalt (commonly called hard asphalt)

625. The main problem associated with wrong practice of conveying concrete is called ____.

Ans. Segregation

626. The specification call for 1:2:3 concrete mix. In one bagger mixer, which of the following amount of water is most likely required for the mix?

Ans. 20 liters

627. A one lane road, with a pavement width of 3.5 meters and shoulder width of 0.5 meter on each side of the pavement, was designed to have sub base and base courses with a combined thickness of 300 mm. If the thickness of the sub base is a 125 mm. the spreading and compaction of the sub base and base course shall be carried out in ____.

Ans. 3 layer over the full width of 4.5 m.

628. As Material Engineer, what is the area of trial section that you would recommend to check the suitability of the materials and efficiency of the equipment to be used for item 310, Bituminous Concrete Surface course?

Ans. about 500 sq. m.

629. Per program of work, the estimated volume of item 201 materials is 7,500 cu. m. How many grading and plasticity test are required based on the DPWH minimum testing requirements?

Ans. 25 grading and 25 plasticity

630. Bituminous concrete mixes contain asphalt by weight equivalent to what?

Ans. 5 to 8%

631. Quality of factory-produced reinforced concrete pipes may be best established through what?

Ans. Test of concrete pipe samples

632. Where do we take additional cores when the measurement of any core taken from the pavement is deficient in thickness by more than 25 mm?

Ans. At no less than 5 meters intervals parallel to the centerline in each direction from the affected location

633. The results of Immersion-Compression test conducted on an asphalt mix sample are as follows:

Dry stability = 2,000 KPa
Wet stability = 1,000 KPa

What is the resulting index of retained strength? Does it meet the specification requirement of item 310 Bituminous Concrete Course?

Ans. Index of Retained strength = $\frac{\text{Wet stability}}{\text{Dry stability}} \times 100$
= $\frac{1,000 \text{ KPa}}{2,000 \text{ KPa}} \times 100$
= 50%

No it does not meet the specification of item 310 since the specification of IRS is 70% minimum while the resulting IRS is only 50%

634. Quality Control testing in a DPWH project is the responsibility of the ____.

Ans. DPWH and Contractor's Materials Engineers

635. As Material Engineer, when is the right time to start sampling and testing aggregates of the course materials?

Ans. before the course materials be incorporated in the project

636. The spot test on asphalt is used to determine what?

Ans. overheating during manufacture

637. Under what climatic condition(s) do we use the lower penetration grades or hard asphalt?

Ans. hot climate

638. Sampling of the freshly mixed concrete should be taken on the ____ batch discharge of the transit mixer.

Ans. middle

639. Soft and unstable soils were encountered during the course of subgrade preparation. If you were the Materials Engineer assigned to the project, what is the most effective and cheaper method that you would recommend to stabilize the soil prior to the construction of the sub base course?

Ans. Geotextiles

640. How many times per layer should a concrete specimen molded in a 6"x6"x21" beam mold be rodded?

Ans. 63 times

641. What is the basis for acceptance or rejection of any construction materials?

Ans. Materials Test Report

642. In testing concrete cylinder sample the load applied should be continuous without shock at a constant rate within the range of ____ per second.

Ans. 0.14 to 0.34 MPa per second or 20 to 50 psi per second

643. During the application of the first half of the anticipated load a ____ rate of loading shall be permitted.

Ans. higher rate

644. Initial rolling in bituminous mixes shall be performed with a ____ roller.

Ans. Pneumatic Tire Roller

645. Final rolling in bituminous mixes shall be performed with a ____ roller.

Ans. 3-wheel or Tandem type steel wheel roller

646. Rolling of bituminous mix shall be discontinued whenever it begins to produce excessive ____ or ____.

Ans. pulverizing of the aggregate or displacement of the mixture

647. Final rolling shall be continued until roller marks are eliminated and a minimum of ____ mass percent of the density of the laboratory compacted specimens prepared in accordance with AASHTO T160 have been obtained.
- Ans. 95 mass percent
648. Sample from compacted bituminous mix shall be neatly cut by saw or core drill. Each sample shall be at least ____.
- Ans. 150 mm. x 150 mm. or 100mm diameter full depth
649. For surface tolerance the surface (bituminous mix) will be check by the use of a ____ at sites selected by the Engineer.
- Ans. 3 meter straight edge
650. What is the sample requirement for paving concrete for every 75 cu. m. or fraction thereof?
- Ans. One (1) set (consisting of 3 pcs.) concrete beam sample
651. In testing bituminous mix sample for Immersion-Compression Stability the sample is soaked in water bath for ____ days.
- Ans. four (4)
652. Given: Wt. of dry aggregate = 950 grams
Wt. of asphalt = 50 grams
- What is the total % of asphalt weight of total mix?
- Ans. Wt. of total mix = wt. of dry agg. + wt. of asphalt
= 950 + 50
= 1,000 grams
- therefore: $(50\text{grams}/1,000\text{grams}) \times 100 = 5\%$
653. Does the percent asphalt in problem above meet the required specification for percent asphalt in bituminous mixes? Why?
- Ans. yes, because standard specification for percent asphalt in bituminous mixes is 5 to 8%
654. If the actual percent asphalt in the problem above is 5% and the percent asphalt in a job-mix 5.6% did it pass the required tolerance? Why?
- Ans. No, because the required tolerance for % asphalt in a job-mix formula is $\pm 0.4\%$
655. Which is not the duty or responsibility of Materials Engineer?
- Ans. quantity of construction materials
656. Which is the duty or responsibility of a Materials engineer?
- Ans. quality of construction materials
657. Problem regarding quality of construction materials shall be reported directly by the contractor's Materials Engineer to the ____.
- Ans. DPWH Materials Engineer
658. How many concrete cylinders are required for a 3,000 pcs? Reinforced concrete pipe culvert?
- Ans. 120 sets (consisting of 3 pcs. Concrete cylinder per set)
659. If the concrete cylinder is not available for a 3,000 pcs. culvert pipes, how many pipes shall be subjected to test?

Ans. 60 pcs. culvert pipes

660. What kind of test that requires calibrated sand in filling the hole in order to determine the volume of hole?

Ans. Field Density Test (FDT)

661. What do you call the test using a rammer and a mold?

Ans. Moisture Density Relation Test or Compaction Test

662. What kind of asphalt has a penetration grade of 0 to 30 mm?

Ans. Blown Asphalt

663. If there is a necessary to add water to the concrete mix in order to increase its workability (provided concrete does not exceed specified slump). How many minutes after the initial of the mixing does adding of water be permitted?

Ans. not exceeding 45 minutes and that water cement ratio is not exceeded

664. What is the percent weight of aggregates (coarse, fine and mineral filler) in the mixture of bituminous concrete surface course?

Ans. 92 to 95%

665. What is the percent weight of asphalt in the bituminous mixture?

Ans. 5 to 8%

666. How can we establish the exact percentage to be used in the bituminous mixture?

Ans. Job-mix formula

667. What are the two types of bitumen that are commonly used in bituminous concrete surface course?

Ans. 1) Tar- is viscous liquid obtained from distillation of coal or wood
2) Petroleum Asphalt - are the product of the distillation of crude oil

668. What are the three major petroleum asphalt product?

Ans. 1) Asphalt cement - produced in various viscosity grades, the most common penetration grade are, 200/300, 120/150, 85/100, 40/50
2) Cut-back asphalt - are asphalt mixed with solvents to reduce their viscosity
3) Emulsified Asphalt - are mixture of asphalt cement and water with emulsifying agent

669. What are the solvents for cut back asphalt?

Ans. 1) gasoline - for rapid curing type
2) kerosene - for medium curing type
3) diesel fuel - for slow curing type

670. Emulsified asphalt is either ____ or ____.

Ans. 1) cationic emulsion - which work better with wet aggregates and in cold weather
2) anionic emulsion - which adhere better to aggregates which have positive charges

671. What is the bituminous material commonly used in item 310?

- Ans. Asphalt cement (AC)
672. What greatly influence the service of an asphalt pavement?
- Ans. grade and quantity of asphalt
673. What influence primarily the grade of asphalt selected?
- Ans. Climatic condition
674. The lower penetration grade or hard asphalt is used in ____.
- Ans. hot climate
675. The higher penetration grade or softer asphalt is used in ____.
- Ans. cold climate
676. How many weeks do a producer of asphalt mix or the contractor shall submit the job-mix formula?
- Ans. three (3) weeks prior to production
677. Who will approve the job -mix formula?
- Ans. DPWH Project Engineer and Materials Engineer
678. The job-mix formula contains provisions on the following:
- Ans. 1) grading
2) percentage and type of asphalt
3) temperature of aggregates and asphalt
4) temperature of mixture upon delivery or at time of compaction
679. What is the primary quality control mechanism for the production of asphalt mixture?
- Ans. Job-mix formula which shall conform to the following ranges of tolerance: Passing No. 4 and larger sieves $\pm 7\%$
Passing No. 8 to No. 100 sieves $\pm 4\%$
Passing No. 200 $\pm 2\%$
Bituminous Materials $\pm 0.4\%$
Temperature of mixture $\pm 10^\circ\text{C}$
680. What is the minimum dry compressive strength of the mixture?
- Ans. 1.4 MPa (200 psi)
681. The mixture shall have a mass percent air voids with the range of ____.
- Ans. 3.0% - 5%
682. What is the Index of Retained Strength of the mixture?
- Ans. 70%
683. The quality control test for asphalt mix are the following:
- of
Ans. 1) Immersion-Compression Test - determines the index of retained strength of the mix
2) Extraction and Grading Test - determines the asphalt content and particles size distribution of coarse and fine aggregates
3) Laboratory density test - determines the degree of compaction

684. What is the correct temperature that the mixture shall be placed as measured in the truck just prior to dumping into the spreader?
Ans. 107°C
685. When tar is used, what is the temperature that the mixture shall be placed?
Ans. 66°C to 107°C
686. When shall the compaction be done in order to attain the required density?
Ans. when the mixture is still hot and workable
687. How to determine the number of passes that would attain the required density?
Ans. trial section
688. What control during the mixing and compaction is of great significance in the strength of the resulting pavement?
Ans. temperature
689. What kind of roller should be used for the initial or breakdown compaction?
Ans. Pneumatic smooth wheeled roller (not less than 10 tons)
690. What is the speed of the roller?
Ans. 5 km/hr.
691. How is the rolling be done?
Ans. It should begin from the side and proceed longitudinally parallel toward the road centerline, each trip overlapping one half the rollers width
692. After the final rolling what should be checked?
Ans. degree of compaction
693. The compacted pavement shall have a density equal to or greater than _____.
Ans. 97% of the laboratory compacted density
694. When is the traffic be permitted to enter the pavement?
Ans. when the pavement has cooled to atmospheric temperature
695. How many samples shall be taken for each full day's operation?
Ans. at least one but not more than three (3) core samples
696. What is the dimension of the samples taken full depth from the finish pavement for thickness determination?
Ans. 150 mm. or 100 mm diameter
697. How is the sample taken from the finished pavement?
Ans. by means of core drill or saw
698. How is the surface level of the finished pavement being check?
Ans. by the use of 3 meters straight edge
699. The variation of the surface from the testing edge of the straight between contacts with the surface shall not exceed ____.

- Ans. 6 mm
700. What determine the thickness and density of compacted asphalt pavement?
- Ans. core test
701. What is the strength requirement (flexural strength) of a concrete when tested by third point loading method?
- Ans. 3.8 MPa (550 psi)
702. What is the compressive strength of the PCCP when rested at 14 days?
- Ans. 24.1 MPa (3500 psi)
703. What is the slump of a workable concrete if not vibrated?
- Ans. 40–75 mm, when vibrated the slump is 10 to 40 mm
704. What kind of form used in item 311?
- Ans. steel forms and a depth equal to the thickness of the pavement
705. How long shall the forms remain undisturbed after concrete pouring?
- Ans. 24 hrs.
706. What is the difference between item 504 and item 505?
- Ans. placement of stone or boulders and the ratio of cement and fine sand
707. What is the ratio of fine aggregates to number of bags of cement for grout in item 504 and 505?
- Ans. for item 504 its one part cement to three part fine sand (1:3)
for item 505 its one part cement to two parts fine sand (1:2)
708. Penetration and ductility test of asphalt are done in ____ temperature.
- | | |
|---------|---------|
| a) 60°C | b) 25°C |
| c) 50°C | d) 10°C |
709. This is the most fluid and lightest of all the grade of Medium Curing Cut-back Asphalt.
- | | |
|-------------|-------------|
| a) MC - 30 | b) MC - 70 |
| c) MC - 800 | d) MC - 300 |
710. The only kind of asphalt that can be used in all bituminous item of work.
- | | |
|-----------------------|---------------------|
| a) Asphalt Cement | b) Cut-back Asphalt |
| c) Emulsified Asphalt | d) All of the above |
711. Retempering of concrete or mortar, which has partially hardened, will be permitted with the addition of cement, aggregate or water.
- | |
|--|
| a) yes |
| b) no |
| c) it depends on the condition of concrete |
| d) check the slump |
712. What is the minimum number of representative cores that shall be taken from each member area of concrete in place that is considered deficient?
- | |
|------------|
| a) 3 cores |
| b) 4 cores |

726. What is item 504 in DPWH specifications?
Ans. Grouted Riprap
727. What is item 505 in DPWH specifications?
Ans. Stone Masonry
728. This is the empirical indicator of the clay fraction of a binder material in a stabilized soil mixture.
- | | | | |
|----|------------------|----|---------------|
| a) | Group Index | b) | Liquid Limit |
| c) | Plasticity Index | d) | Plastic Limit |
729. When water is added to the mixture to bring the slump within required limit, it shall be injected into the mixer under such pressure and direction of flow that the requirement for uniformity and desired slump of concrete are met. The drum shall be turned on additional ____ revolution or more.
- | | | | |
|----|----------------|----|----------------|
| a) | 10 revolutions | b) | 45 revolutions |
| c) | 30 revolutions | d) | 15 revolutions |
730. Job-mix formula submitted for approval shall propose definite single values for, except ____.
- | | |
|----|--|
| a) | % of aggregate passing each specified sieve |
| b) | temperature of the mixture delivered on the road |
| c) | percent of bituminous materials to be added |
| d) | mixing time of the mixture in the batching plant |
731. For small contract project of the DPWH located in remote areas where the quality of cement to be used is too small and where there are no accessible testing laboratory to check the quality of materials, exception in the requirements for performance of laboratory test in cement prior to use may be allowed provided that the trial mixes be done for every ____.
- | | |
|----|--|
| a) | deformation characteristic of soil |
| b) | engineering properties of the underlying soils |
| c) | arrangement of soil strata |
| d) | field density of soil |
732. How many days shall the Engineer be notified in advance of the date that the treating process will be performed in order that the untreated lumber, the treatment process and the finished treatment timber may all be inspected?
- | | | | |
|----|---------|----|---------|
| a) | 15 days | b) | 10 days |
| c) | 21 days | d) | 7 days |
733. What test determined the crushing of strength of aggregates to ensure that coarse aggregates will not degrade to smaller ones?
- | | | | |
|----|----------|----|-----------------|
| a) | Abrasion | b) | Grading |
| c) | Density | d) | Plasticity test |
734. A one-lane road, with of 3.5 meters and shoulder width of 0.5 meter on each side of the pavement was designed to have sub-base and base coarse with a combined thickness of 300mm. If the thickness of the sub-base is 125mm, the spreading and compaction of the sub-base and base coarse shall be carried out in ____
- | | |
|----|---------------------------------------|
| a) | 3 layers over the full width of 4.5 m |
| b) | 2 layers over the full width of 4.5 m |
| c) | 3 layers over the full width of 3.5 m |
| d) | 2 layers over the full width of 3.5 m |
735. Weakened joints on concrete pavement should be sawed ____

- a) Immediately after the pavement surface has been brooms finished
- b) Sawing of joints shall commence as soon as the concrete has hardened sufficiently, usually within 24 hours
- c) Before the pavements opened to traffic usually 14 days from the day of pouring

736. What depth of weakened joint should at all times not be less than ____

- a) 50 mm
- b) 60 mm
- c) 70 mm
- d) 75 mm

737. Cores are taken from a concrete pavement and tested for ____

- a) Strength verification of concrete in place that is considered adequate
- b) Flexural strength determination
- c) Cement content determination
- d) Thickness determination

738. In accepting completed concrete pavement poured by lane thickness ____ determination shall come out by obtaining one core from every

- a) 100 meters
- b) 200 meters
- c) 500 meters
- d) 1000 meters

739. If the strength of Item 311 control specimens does not meet the requirements by not more than 25% and it is not feasible to obtain cores from the structure due to structural consideration:

- a) The strength of in-place concrete will be determined by rebound a hammer
- b) Pavement of the concrete will be made at an adjusted price
- c) No pavement will be made on the concrete represented by the failed specimen
- d) The contractor will be required to remove and replace the concrete represented by the failed specimens at his own expense.

740. The flexural strength requirements of item 311 is

- a) 500 psi
- b) 550 psi
- c) 3000 psi
- d) 3500 psi

764. The number of sets of beam test specimens that should be taken from 700 sq.m. of concrete pavements, 0.23 cm thick poured on a single day should be:

- a) 2
- b) 3
- c) 6
- d) 9

765. Three cores taken from a pavement section poured on a single day were tested to have strength as follows:

Core		
1	3650	
2	3000	86% of 3500 psi
3	2450	70% of 3500 psi
Average	3033	87% of 3500 psi

If the required strength is 3500 psi concrete are represented by the cores

- a) Maybe accepted, subject to price adjustment
- b) Is considered adequate
- c) Is considered inadequate
- d) May be recorded at the initiative of the implementing office

766. Forms for concrete pavement shall remain in place undisturbed for not less than

- a) 24 hours after concrete pouring
- b) The curing period

779. The grade designation of a reinforcing steel bars is also its
- a) Nominal area
 - b) Yield strength
 - c) Tensile strength
 - d) Unit mass
780. Which of the following is not an engineering property of granular soils?
- a) Good load bearing qualities
 - b) Drain readily
 - c) Susceptible to strength or volume change due to change in water content
 - d) Comparatively incompressible due to static load
781. Peat is an example of
- a) Inorganic soil
 - b) Organic soil
 - c) Selected borrow
 - d) Embankment material
782. The specific irregularity tolerance for item 201 is
- a) 5 mm
 - b) 10 mm
 - c) 15 mm
 - d) 20 mm
783. The depth of the weakened plane joint with a width of 6 mm is
- a) 40 mm
 - b) 45 mm
 - c) 50 mm
 - d) 60 mm
784. Which of the following is not the job for a materials engineer?
- a) Submits samples for testing
 - b) Prepares design mix
 - c) Checks job mix formula
 - d) Surveys and measures line and grade at site
785. The proper way of sampling of aggregate base course in the laboratory if the splitter is not available is to
- a) Scoop for samples in the middle of the sack
 - b) Spread the samples on the ground and take the samples in the middle
 - c) Spread the samples on the ground and divide the samples into four and take two opposite parts from it.
 - d) Scoop for samples at the side of the sack
786. Given : Original weight of sample: 5000g
 Sample retained on No. 12 sieve: 4178g
- What is the abrasion loss?
- a) 16.42%
 - b) 16.44%
 - c) 19.67%
 - d) 19.76%
787. Which is true on conducting of a slump test?
- a) Consistency of mixture is checked by the slump test
 - b) It is simple, but important, n indicator of water – cement ratio
 - c) Slump cone is used together with tamping rod and ruler
 - d) All of the above
788. Reinforced concrete pipe culverts are tested for compressive strength to produce _____ crack.
- a) 0.20 mm
 - b) 0.25 mm
 - c) 0.30 mm
 - d) 0.35 mm
789. Item 310 mixture shall have an index of retained strength of not less than
- a) 66
 - b) 70

- c) 97 d) 107

790. Item 310 mixture shall be delivered to the paver at a temperature not less than

a) 66°C b) 97°C
c) 107°C d) 110°C

791. In the asphaltting of a two lane, 500 meter road which lasted for three days, how many core samples should be taken from the pavement if no sample was taken at the end of each day's operation?

a) 3 b) 5
c) 6 d) 10

792. Which of the following provisions is not included in the job - mix formula?

a) Mixing time b) Grading of aggregates
c) Asphalt cement d) Temperature at delivery

793. A bituminous mix subjected to immersion - compression test yielded a dry stability of 2.0 kg and a wet stability of 1.5 kg. The index of retained strength of the mix is

a) 0.5 b) 1.33
c) 33 d) 75

794. In item 310, bitumen content, expressed in terms of percentage by mass of aggregate, shall be

a) 3 to 5% b) 4 to 7%
c) 5 to 8% d) 6 to 8%

795. In the asphaltting of a 3 - meter wide, 200 meter long road lane, a core was obtained and tested to have a thickness of 5 cm. and a density of 2.25 gm/cm³. The quantity of bituminous mix to be paid, in tonnes, is

a) 65.5 b) 67.5
c) 655 d) 675

796. After final rolling, the compacted asphalt pavement shall have a density equal to, or greater than % of the laboratory compacted density.

a) 90 b) 95
c) 97 d) 98

797. Which of the following is an engineering property of line grained soil?

a) hard to drain
b) Susceptible to volume change under vibratory load, if not adequately compacted
c) Stable when subjected to load
d) Settlement is high

798. The dividing size between coarse and fine aggregates used in asphaltting is

a) No. 4 b) No. 8
c) No. 16 d) No. 30

799. The objective of designing the asphalt mixture is determination of

a) Index of retained strength b) Penetration grade of asphalt
c) Temperature of mix d) Production control

800. The sample of reinforcing steel bars for every 10,000 kgs shall be

a) 1 meter long b) 2 meters long
c) 4 meters long d) 6 meters long

801. Given: Specific gravity = 2.305

Wet compressive strength	=	1000 kPa
Dry compressive strength	=	1400 kPa

What is the index of retained strength value?

- | | | | |
|----|-------|----|-------|
| a) | 70.1% | b) | 71.4% |
| c) | 40% | d) | 140% |